

claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\workflows\wf_cb1d52fa-fed\agent-ad91a8ddb06f32646.jsonl`  
- SHA-256: `b7162c1e317f7a2607ecc49ff36084cf016747222eebcf406613115fa7d755a0`  
- Source modified: `2026-06-14T08:27:33+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_cb1d52fa-fed`  
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`
```

Transcript

1. user (2026-06-14T08:24:06.518Z)

THE PIVOT under study = "PERSONALIZATION-FIRST": instead of one universal model for all users, ship a BATTERIES-INCLUDED baseline that CONTINUALLY SELF-OPTIMIZES to a SINGLE user's own footage. As the user's corpus grows (10-15 videos), the on-device model + PER-USER CUE SELECTION (driven by the EXISTING LOVO / A-B experiment harness run on the user's OWN videos, with the user's CORRECTION LOOP as the per-user label source) becomes heavily tuned for THAT user ? deep analysis of THEIR rallies + shot types (serves, smashes, drives, drop shots). Per-user decisions like "the motion/person cue gives no F1 win for THIS user -> default it OFF for them." The model is optimized for the user's videos, NOT others (wins may hold anecdotally on other footage but weaker). Periodic "BATTERIES UPGRADES" improve the shipped baseline so it needs merely <5 videos to reach superior quality. Two user archetypes: (a) an economically-generous hobbyist who records sessions some days; (b) a professional with 100+ hours of their own video. CONTRAST = "generalization-first" (build one model that works on everyone's videos). The owner believes the harness + eval + A/B were DESIGNED to enable this roadmap.

DIMENSION: THE DEEP-ANALYSIS / SHOT-TAXONOMY feasibility (the harder part). The user wants per-user detection of serves, smashes, drives, drop shots, clears etc. ? finer-grained than rally detection. Analyze: cold-start for SHOT TYPES per user is harder than for rallies (needs per-user shot labels via the correction loop; class imbalance; a hobbyist may not label enough smashes in 10-15 videos); is "heavily optimized by 10-15 videos" realistic for shot taxonomy vs just rally/highlight detection? Suggest a phasing (rally+highlight first, shot-types as a later layer once per-user labels accumulate). Read docs/HOW_RALLY_DETECTION_WORKS.md for the current scope. Return pros/cons + a realistic phasing.

2. assistant (2026-06-14T08:24:09.647Z)

I'll analyze the deep-analysis / shot-taxonomy feasibility dimension of this personalization-first pivot. Let me start by reading the key doc that defines current scope.

3. user (2026-06-14T08:24:10.602Z)

```
1      # How rally detection works (the 2-layer mental model)
2
3      A short conceptual map of how we go from raw video to rally `[start, end]` segments ?
4      and, crucially, **which layer our annotation can improve.**
5
6      ```
7      video ??? frames ??? [DETECTION] ??? shuttle (x,y,visible) per frame
8              ?
9      (WASB CNN, per-frame trajectory
10     frozen weights)
11              ?
12              velocity signal (how fast the shuttle moves)
13              ?
14     [SEGMENTATION] ??? the 3 knobs we tune
```

```

15                                     ?
16                                     ?
17                                rally windows [start, end] ??? IoU vs golden
labels
18    ``
19
20    ## Two layers ? keep them separate in your head
21
22    **1. DETECTION ? "where is the shuttle, this frame?"**
23    WASB is a CNN that, for each frame, outputs a heatmap ? the shuttle's `(x, y)` + a
24    visibility/confidence. Its weights were learned (by gradient descent, on shuttle-
coordinate
25    labels) and we use them **frozen**. We chose a *shuttle* detector on purpose: "shuttle
in
26    flight" is the signal for "a rally is happening," and it's **camera-agnostic** ? unlike
27    player-motion, which fooled the old heuristic on broadcast cameras that pan with the
players.
28    (Your videos are static tripod, so player-motion would also work ? but shuttle-motion is
the
29    robust general bet.)
30
31    **2. SEGMENTATION ? "when is a rally?"**
32    From the per-frame `(x, y)` we compute the shuttle's **velocity** (displacement between
frames).
33    Then three knobs turn that into rally intervals:
34    - **`velocity_thresh`** ? a frame is "active" only if the shuttle is moving faster than
this
35    (in flight, not lying on the floor / being picked up).
36    - **`merge_gap`** ? active frames within this many seconds get joined into one rally
(bridges
37    brief gaps: a shuttle momentarily occluded, or the beat between two hits).
38    - **`min_window_duration`** ? throw away windows shorter than this (blips / false
flickers).
39
40    Output: a list of `[start, end]` windows = the rallies.
41
42    ## Where the annotation fits (the key point)
43    Our Roora/golden labels are **rally `[start, end]` times** ? they say *when* rallies
are, not
44    *where the shuttle is*. So they can't teach WASB's detector (that needs `(x,y)` labels).
They
45    **can** grade and tune the **segmentation layer**: we score predicted windows against
the golden
46    rallies with **temporal IoU** (how much they overlap), and pick the 3 knobs that
maximise it ?
47    **calibrate on some rallies, measure on held-out ones** so the number is honest.
48
49    ## Where sparks could land (open questions ? not commitments)
50    Things worth a shower-thought; any of these could beat 3 hand-tuned thresholds:
51    - **Richer features than raw velocity:** acceleration, trajectory *continuity* (a real
rally is a
52    smooth arc that survives occlusion), shuttle *height/arc*, or **direction reversals**
(each
53    reversal ? a shot ? ties to `shots_count`!).
54    - **A tiny learned segmenter** instead of 3 thresholds: feed the trajectory features to
a small
55    model trained on the golden rally labels (still the segmentation layer, just smarter).
56    - **Hysteresis** (two thresholds: enter-rally high, stay-in-rally low) ? robust to mid-
rally dips.
57    - **Rhythm/periodicity:** rallies have a hit cadence; dead time doesn't.
58    - **Fusion:** combine shuttle-motion (WASB) with player-motion for static cams. ?
designed in
59    [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md) (one rally-fusion layer;
the person cue is
60    ADR-007's parked #2 and `rally_gate.py`'s named-but-unbuilt "Gate B").

```

61 - ****Per-video calibration as a *feature*:**** "watch a few labeled rallies ? auto-tune to
this
62 court/lighting ? segment the rest sharply" ? exactly your earlier idea, at the right
layer.
63
64 The whole point of per-video calibration: a handful of ***your*** labeled rallies nudges
these knobs
65 to ***your*** court/camera, and we report the IoU lift on the rallies it never saw.
66

4. user (2026-06-14T08:24:13.018Z)

fatal: Cannot fast-forward to multiple branches.

5. assistant (2026-06-14T08:24:16.003Z)

The git pull failed but that's a pre-existing repo state issue (multiple branches), not something I should force. I'll proceed read-only on local state. Let me read the key supporting docs to ground my analysis of the shot-taxonomy feasibility.

6. user (2026-06-14T08:24:18.543Z)

ANALYZERS_AND_RECONCILERS.md
BACKLOG.md
CODE_MAP.md
COMMERCIALIZATION.md
DEFERRED_HARDENING_PLAN.md
EVALUATION.md
EXECUTION_POLICY.md
EXPERIMENT_HARNESS.md
FIELD_VISIT_ANSWER_SHEET.md
FIELD_VISIT_PLAYBOOK.md
GOLDEN_SET_IMPLEMENTATION.md
GOLDEN_SET_PHASE1_VIDEOS.md
GOLDEN_SET_VENDORING.md
HOSTING_PLAN.md
HOW_RALLY_DETECTION_WORKS.md
I18N_PLAN.md
MULTI_SIGNAL_FUSION_PLAN.md
NEXT_STEPS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PER_VIDEO_OUTPUT_LAYOUT.md
PLATFORM_ARCHITECTURE.md
PMF_FIELD_SIGNALS.md
PMF_MARKET_RESEARCH.md
PRODUCT_FIELD_ASSOCIATE_FLYER.md
QUALITY_ITERATIONS.md
README.md
ROADMAP.md
ROORA_LABELING_GUIDELINES.md
STORAGE_SHARING_MODEL.md
TRACKNET_WSL_SETUP.md
UI_ALPHA_EXECUTION_PLAN.md
UI_PROPOSAL.md
VIDEO_LOCALITY_MODEL.md
VIDEO_RECORDING_GUIDELINES.md
archives
mockups
---ROORA---
EXPERIMENT_HARNESS.md
MULTI_SIGNAL_FUSION_PLAN.md
QUALITY_ITERATIONS.md
ROORA_LABELING_GUIDELINES.md

7. user (2026-06-14T08:24:18.733Z)

```
1      # Multi-signal rally fusion ? add a human (person) cue alongside the shuttle
2
3      **Status:** **P1 DONE** (2026-06-13) ? the person cue is extracted to
4      `backend/pipeline/detectors/person_detector.py` (`PersonDetector`), consumed by
5      `yolo_hybrid` with no
6      behavior change. **P2?P4 remain owner-gated.** Phases below are sequenced but each is a
7      separate PR.
8      **Owner decisions captured** in [S9](#9-open-decisions-for-the-owner).
9
10     > **Companion docs:** the 2-layer mental model is
11     [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md);
12     > the modular-fusion architecture this builds on is **ADR-007**
13     ([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md));
14     > the permissive-dependency rule is **ADR-002 / ADR-005** ; the audio cue (the #1 next
15     cue, already
16     scaffolded) is [archives/research/AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUDIO_RALLY_DETECTION_PLAN.md).
17     > The vendor labels that gate the learned path come from
18     [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md)
19     > and the labeling spec in [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md).
20     > **How each cue here is added/tested without production regression** ? A/B, shadow
21     eval, and the
22     > promote-only-on-lift discipline ? is [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)
23     (every P1?P4 cue
24     > below lands flag-gated, default OFF, promoted only on measured LOVO lift).
25
26     ---
27
28     ## 1. Context ? why this, why now
29
30     Rally detection today runs **one cue at a time**. Each segmenter independently turns a
31     single signal
32     into rally windows:
33
34     | Segmenter | Signal | File |
35     |---|---|---|
36     | `wasb_hybrid` / `tracknet_hybrid` | shuttle velocity (frozen WASB/TrackNet trajectory)
37     | `backend/pipeline/detectors/tracknet_runner.py` |
38     | `yolo_hybrid` (the current `default_segmenter`) | person velocity (torchvision +
39     ByteTrack) | `backend/pipeline/segmenters/yolo_hybrid.py` |
40     | `motion` | raw pixel motion | `backend/pipeline/segmenters/motion.py` |
41     | `gemini` | a cloud VLM watching the whole clip | `backend/providers/gemini.py` |
42
43     **There is no fusion.** The owner's directive: make the detector **use the shuttle *and*
44     the humans
45     together** ? and be ready to fold in more signals (audio, later pose) ? delivered
46     **MIT / Apache-2.0 / BSD-clean** so we can ramp commercially without a licensing
47     rethink.
48
49     This is **not greenfield** ? it operationalizes work the repo already blessed:
50
51     - **`HOW_RALLY_DETECTION_WORKS.md` open question:** **"Fusion: combine shuttle-motion
52     (WASB) with
53     player-motion for static cams."
54     - **ADR-007** already designed a **modular "rally layer" that fuses cues *outside* the
55     frozen WASB
56     model, with **audio as the #1 next cue and person/pose parked as #2**. This plan picks
57     up the parked
58     person cue and generalizes the fusion seam so future cues are "register a producer,"
59     not "re-touch
60     the segmenter."
61     - **`backend/pipeline/detectors/rally_gate.py` already exists** ? the net-crossing gate,
62     explicitly
```

```

45     labelled in its own docstring as "Gate A in the over-fire fix spectrum (**B = player-
stance state
46     machine, future)"." The person cue in this plan "is that named-but-unbuilt Gate
B."
47
48     ### The owner's "held-shuttle" false positive ? what's actually going on
49
50     The owner observed a "rally" that was just a player "holding / flicking the shuttle in
hand" between
51     points. This is the *exact* failure `rally_gate.py` was written to kill:
52
53     > "WASB over-detects: when a player flicks/tosses the shuttle back to the opponent for
service after
54     > losing a point, that single trajectory looks like a rally. The discriminator is
structural: a real
55     > rally has the shuttle crossing the net MULTIPLE times; a single service feed crosses
~once."
56
57     "The structural fix already exists and is validated offline ? but it is NOT wired into
the production
58     runtime path." `rally_gate.apply_gate(..., min_crossings=2)` is called only from the
"eval drivers"
59     (`distill_local.py`, `eval_partial_labels.py`, `export_wasb_segments.py`,
`calibrate_local.py`); there
60     is "no `min_crossings` knob in `IndexingConfig`" and the live segmenters don't apply
it. So the
61     single cheapest, highest-confidence win in this whole plan is "P2's first step: fold
the existing
62     Gate A into production." Everything else (the person cue / Gate B, the learned fusion)
is robustness
63     on top of that.
64
65     ### Why humans help ? and the honest risk
66
67     The beachhead videos are "static tripod" (Bangalore academies,
[PMF_MARKET_RESEARCH.md](PMF_MARKET_RESEARCH.md)),
68     so player presence/motion is a usable cue. ADR-001's warning was specifically about
"broadcast pan"
69     cameras, where player-motion fooled the old heuristic ? *that failure mode does not
apply to a fixed
70     court cam.* But we keep "the shuttle as the camera-invariant primary" so the general
bet stays robust
71     on other camera types. Humans then add:
72
73     - "Precision" ? reject shuttle-on-floor / warm-up knock-ups / held-shuttle feeds when
the court
74     isn't occupied by ?2 players in a play posture.
75     - "Recall" (via the learned classifier) ? features that help bridge shuttle
occlusion/blur gaps.
76
77     ---
78
79     ## 2. Core architecture ? one rally-fusion layer, cues are swappable producers
80
81     Formalize ADR-007's diagram. Every cue is a "black box that emits a time-aligned
signal" (per-frame
82     activity and/or per-window features); "fusion happens in *our* layer, never inside a
frozen model."
83
84     ```
85     frames ?? WASB (frozen, MIT)      ?? shuttle trajectory ???
86     frames ?? person detector (ours)  ?? tracks + boxes ??????
87     audio ?? hit detector (ours)      ?? hit events ???????????? RALLY LAYER (ours) ??
rallies
88     [frames ?? pose/stance (ours; parked #3) ?? stance ???????? feature + decision fusion

```

```

89     ``
90
91     - **Shuttle = primary, camera-invariant.** Windows still **seed** from
92       `TrackNetRunner.trajectory_to_action_windows()` (`tracknet_runner.py:234`). Other cues
93       *enrich and
94       adjudicate* shuttle-seeded windows; they do **not** replace the shuttle seed. Person
95       runs only over
96       **shuttle-seeded candidate windows + padding**, not the whole video ? which bounds the
97       added cost.
98       - **A small `RallyCue` contract.** Each producer yields `(frame_idx ? features)` + per-
99       window stats.
100       Adding a cue = register a producer; the existing `SegmenterRegistry`
101       (`segmenters/base.py:35`) is the
102       template for the pattern. Keep the single-cue segmenters registered as fallbacks.
103
104     ---
105
106     ## 3. The two fusion modes (both ? per the owner decision)
107
108     ### 3.1 Decision-level gate (safe baseline, ships first, no labels needed)
109
110     ``
111     rally_active = shuttle_motion ? (?? persons in court zone ? recent person track)
112     ``
113
114     The `? recent-track` clause tolerates 1?2 frame person-detector misses so the AND-gate
115     can't cause a
116     recall cliff. This composes with the **existing net-crossing Gate A**:
117
118     ``
119     keep window ? net_crossings ? 2                (Gate A ? EXISTS in rally_gate.py, eval-
120     only today)
121     ? (?? in-court players ? recent track)    (Gate B ? the person cue, NEW)
122     ``
123
124     Gate A kills the service-feed / held-shuttle case structurally; Gate B adds court-
125     occupancy evidence
126     for the cases a single trajectory can still fool (e.g. a knock-up that does cross the
127     net). No training
128     data required ? this is the label-free fusion that ships before any golden labels exist.
129
130     ### 3.2 Feature-level (primary, the learned path)
131
132     The cues emit a **small set of aggregated, scale/translation-invariant per-window
133     features ? NOT raw
134     coordinates.** Raw pixel coordinates would memorize *this* court/camera and fail to
135     generalize off a
136     still-small labeled set; counts / ratios / zone-membership / two-sidedness transfer.
137
138     **What already feeds the classifier** (`distill_local.py:40`, `FEATURE_NAMES`):
139     `duration, peak_velocity, mean_velocity, active_ratio, net_crossings` ? note
140     **`net_crossings` is
141     already a feature** (computed via `rally_gate.gate_windows`).
142
143     **Person features to add (~4, deliberately few):**
144
145     | Feature | Meaning | Rally signal |
146     |---|---|---|
147     | `players_in_court` | median count of persistent in-court tracks | ? 2 (singles) / 4
148     (doubles); 0?1 = dead time |
149     | `two_sided_motion` | are moving players on **both** net-halves? | rally = two-sided;
150     one person feeding = one-sided |
151     | `mean_player_motion` | aggregate normalized player velocity | rally = sustained;
152     resting/standing = low |
153     | `in_court_ratio` | fraction of frames with ?? in-court players | track stability vs

```

```

flicker |
139 | `shuttle_player_colocation` | fraction of frames the shuttle sits inside/adjacent to a
person box (hand region) | **held = high; in-flight = mostly free space** ? the direct held-
shuttle discriminator |
140
141 These join the shuttle features in the candidate `stats` JSON
142 (`database.py::add_candidate_segment`) and the feature vector of the **existing
distilled classifier ?
143 the tiny numpy-only L2 logistic regression in `backend/eval/classifier.py` (ADR-004),
NOT a new net.**
144 It scores each candidate window `P(real rally)` and replaces the Gemini call at runtime.
145
146 **What the model learns:** the **weights** on those few features ? e.g. down-weight a
shuttle-motion
147 window with 0?1 in-court players / one-sided motion / high shuttle-colocation (warm-up,
floor shuttle,
148 held-shuttle feed); up-weight ?2 players, two-sided, sustained motion, shuttle in free
flight.
149
150 **Small-N discipline (non-negotiable):**
151 - Keep to **~4 person features** ? more features on a still-small labeled set overfit;
that's exactly
152 why the model stays a logistic regression, not a net.
153 - Use the **leave-one-video-out (LOVO)** protocol that `distill_local.py` **already
implements**: train
154 on the other videos' candidates, predict the held-out *video*, score vs its labels ?
never a
155 held-out *window* from the same video (windows in one video are correlated; that
flatters the score).
156 - Training labels today are **Gemini silver labels** (distilled offline, ADR-004/007 ?
Gemini is a
157 teacher, never a runtime dependency, never a training-data *source* for owned
weights); **golden
158 labels are the honest measuring stick**, and as the Roora golden corpus grows they can
also become
159 training labels. Until the labeled set is larger, treat the learned weights as
160 **calibration / direction-check, not "a fully general model"*** ? the decision-level
gate (§3.1) is
161 the shippable, label-free fusion in the meantime.
162
163 > *(Gated on golden labels existing ? the same blocker ADR-007 flags for audio. See
164 > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md).)*
165
166 ---
167
168 ## 4. Shuttle-state: in-flight vs in-hand (the misclassification, mapped to real code)
169
170 The "shuttle moved" seed conflates a shuttle **in flight** with a shuttle **carried in
hand**. The
171 discriminators, and where each lives:
172
173 | Signal | In-flight | Held / static | Status |
174 |---|---|---|---|
175 | `net_crossings` | ? 1 (usually several) | ~0 (held) / ~1 (single feed) | **EXISTS**
(`rally_gate.py`); eval-only ? wire to prod |
176 | `shuttle_player_colocation` | low (free space) | high (near a hand) | **NEW** (needs
the person cue) |
177 | `direction_reversals` | several (racket contacts) | ~0 | derivable from trajectory;
partial overlap with crossings |
178 | `peak shuttle speed` | high | walking speed | derivable from existing velocity |
179
180 So the in-flight-vs-held fix is **mostly Gate A (already built) + the new colocation
feature from the
181 person cue**. **No new label column is needed** ? the golden labels already start a
rally at the

```

```

182     *serve-contact frame* and treat all pre-serve / held-shuttle time as dead time (see
183     [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md)), so held-shuttle moments
are negatives
184     by construction and are the ground truth that proves the detector wrong.
185
186     ---
187
188     ## 5. Adversarial mitigations (integrating safely)
189
190     - **Court-region mask** ? polygon around the court; drop persons outside it (spectators,
coaches,
191     adjacent courts). Config-supplied per camera; ~1 ms. Source: a manual per-camera
polygon first;
192     the **4 court corners + net-line** that Roora labels once per video (see the labeling
doc) build
193     this mask **and** the net split that `two_sided_motion` / `net_crossings` need.
194     - **Player-count constraint** ? expect **2 (singles) / 4 (doubles)** persistent in-court
tracks;
195     suppress transient/extra detections. The `singles_or_doubles` manifest flag sets the
expectation.
196     - **Temporal association** ? reuse `supervision` **ByteTrack** (already in
`yolo_hybrid`); reject
197     singleton detections; smooth across frames.
198     - **Multi-court rejection** ? proximity to the primary court region (homography
optional, later).
199     - **Domain shift (COCO frontal ? side/overhead court)** ? start **box-only** (robust);
keep the
200     detector swappable. Fine-tuning a person detector is **parked** ? we have no in-house
person labels,
201     and the **paid-LLM/training guardrail + minors-exclusion** apply to any future person-
training data.
202     - **Fusion-degradation guard** ? shuttle stays primary; person is gate + feature, never
the sole
203     trigger; the single-cue segmenters remain registered fallbacks.
204
205     ---
206
207     ## 6. Box vs pose
208
209     **Box-first.** Occupancy + count + two-sided motion + shuttle-colocation is enough for
both the gate
210     and the features, and box detection is the cheap, robust win. **2D pose is parked behind
the same
211     `RallyCue` interface** ? pose (MediaPipe Pose / RTMPose, both Apache-2.0) buys serve-
ready stance
212     gating and shot cues later, at the cost of a second model. This matches ADR-007 parking
pose as the
213     #2/#3 cue.
214
215     ---
216
217     ## 7. Performance
218
219     The person detector is the **existing in-process torchvision model** run at **frame-skip
3?5**
220     (`yolo_frame_skip` default 3) with track interpolation between sampled frames. The
shuttle keeps
221     running in its separate WSL process ? the two are **separate processes**, so a "shared
backbone" isn't
222     free; budget realistically as the existing `yolo_hybrid` cost minus skip savings. Person
runs **only
223     over shuttle-seeded windows + `ai_handoff_padding`**, not the whole video, which bounds
the cost.
224
225     ---

```



```

226
227     ## 8. Licensing (the guardrail table ? all green)
228
229     Every dependency stays **MIT / Apache-2.0 / BSD** (ADR-002 / ADR-005; `ultralytics` was
dropped for
230     AGPL-3.0).
231
232     | Cue / model | Code | Weights / data | Verdict |
233     |---|---|---|---|
234     | Shuttle: WASB-SBDT, TrackNetV3 | MIT | frozen badminton weights | ? (ADR-001/002) |
235     | Person (default): torchvision Faster R-CNN / RetinaNet / FCOS | BSD/MIT | COCO-
pretrained | ? already in repo (`_DETECTOR_FACTORIES`) |
236     | Person (escalation): **YOLOX**, **RT-DETR / RT-DETRv2** | Apache-2.0 | COCO /
Objects365 | ? |
237     | Pose (parked): **MediaPipe Pose**, **RTMPose / MMPose** | Apache-2.0 | COCO-structure
| ? |
238     | Tracking: `supervision` ByteTrack | MIT | n/a | ? already in repo |
239     | **Banned** | **Ultralytics YOLOv5/8/11 (AGPL-3.0)** . **MonoTrack (Adobe, non-comm)**
. **YOLO-NAS (Deci, non-comm)** | ? | ? |
240
241     **COCO caveat to record:** COCO *annotations* are **CC BY 4.0** (commercial OK *with
attribution*);
242     COCO *images* are Flickr-ToS (per-image). Pretrained-**weight** use is clean; record the
attribution.
243
244     ---
245
246     ## 9. Open decisions (for the owner)
247
248     1. **Court-region mask source** ? manual per-camera polygon (simple, ship first) vs auto
court
249     homography/line detection (later). **Recommend manual first**, fed by Roora's court-
corner labels.
250     2. **Person detector default** ? keep torchvision (zero new deps, BSD/MIT) vs adopt
YOLOX/RT-DETR
251     (Apache, faster) now. **Recommend torchvision first**, escalate only if eval shows a
speed/accuracy gap.
252     3. **Pose now or parked** ? **recommend parked** behind the interface (box-first).
253     4. **Golden labels** ? feature-level fusion (P3) is **blocked on golden rally labels**;
the source is
254     the VLC `rally-annotator` + the Roora vendor pilot
([GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md)).
255
256     ---
257
258     ## 10. Phased execution (owner-gated; ONE PR per slice, off `master`)
259
260     - **P0 ? this design doc** +
[ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md), indexed in
261     `docs/README.md`, cross-linked from `HOW_RALLY_DETECTION_WORKS.md` and ADR-007. *(this
deliverable;
262     docs-only, no code.)*
263     - **P1 ? extract the shared person cue:** ? **DONE (2026-06-13).** Lifted the
torchvision detector +
264     ByteTrack + velocity out of `yolo_hybrid` into
`backend/pipeline/detectors/person_detector.py`
265     (`PersonDetector`); `yolo_hybrid` now orchestrates only and delegates to `self.person`
(no behavior
266     change ? regression-safe, proven by the unchanged process_video output tests). New
267     `tests/test_person_detector.py`; suite +1 green; person_detector is type-clean (left
mypy quarantine
268     for the moved code). The fusion segmenter (P2) reuses this same producer over shuttle-
seeded windows.
269     - **P2 ? the FusionSegmenter (Gate A + Gate B + person-feature persistence):** ? **DONE
(2026-06-14).**

```

```

270     Shipped `backend/pipeline/segmenters/fusion_hybrid.py::FusionSegmenter` (registry
`fusion_hybrid`,
271     **default-OFF** ? `default_segmenter` unchanged), subclassing the trajectory engine
via two identity
272     hooks (`_enrich_candidate_stats`, `_select_for_handoff` ? wasb/tracknet stay byte-
identical, adversarially
273     verified). It seeds windows from the shuttle substrate, **always persists
`net_crossings`** (Gate A signal),
274     and ? when `indexing.fusion.cue_flags.person` is on ? runs the person cue over each
window's frames
275     (`PersonDetector.detections_per_frame`) and persists the ~5 person features
(`fusion_features.py`,
276     pure + unit-tested). Served gating is optional + default-OFF: `fusion.min_crossings`
(Gate A) /
277     `fusion.min_players` (Gate B) skip sub-threshold windows from the handoff while
**persisting the full
278     superset** (offline substrate intact). Court mask = optional `fusion.court_polygon`
(None ? count-only).
279     Lift to be measured via the harness; held-shuttle-drop + person-feature values are
GPU/owner-side.
280     - **P3 ? feature-level (learned) fusion:** add the person features (incl.
`shuttle_player_colocation`)
281     to `distill_local` / `classifier`; recalibrate IoU on the golden set; report **LOVO
lift on held-out
282     videos**. *(Gated on golden labels.)*
283     - **P4 ? additional cues:** wire **audio** (ADR-007's #1, already scaffolded in
284     `backend/pipeline/audio/hit_detector.py`) and later **pose**, through the same
`RallyCue` interface.
285     Each is add-a-producer, not a refactor.
286
287     ---
288
289     ## 11. Verification
290
291     - **Offline, runnable in this dev container (no GPU/torch/cv2):** unit tests for the
fusion gate logic
292     and `RallyCue` aggregation (court-mask filter, player-count constraint, `? recent-
track` tolerance)
293     and the shuttle-state signals (net-crossing count ? extends the existing `rally_gate`
tests ? and
294     person-colocation) with **synthetic trajectories + person boxes**. A key-license
assertion test
295     (no `ultralytics` import; detector factories stay torchvision/Apache).
296     - **Machine-side (owner, GPU/WSL ? the full pipeline can't run in this container):** run
the
297     `FusionSegmenter` on a real static-tripod clip; confirm **held-shuttle / warm-up /
spectator false
298     positives drop** vs shuttle-only (this should reproduce the owner's observed bug and
show it fixed ?
299     start by confirming Gate-A-in-production alone fixes it); **LOVO IoU lift on the
golden set** for the
300     learned path.
301     - **Docs (P0):** links resolve; both docs indexed in `docs/README.md`; this doc cross-
linked from
302     `HOW_RALLY_DETECTION_WORKS.md` (the fusion open-question) and ADR-007.
303

```

8. assistant (2026-06-14T08:24:22.627Z)

Now let me read the experiment harness (the LOVO/A-B machinery the pivot depends on) and the Roora labeling guidelines (to understand what labels the correction loop / vendors actually produce ? critical for whether shot-type labels exist).

9. user (2026-06-14T08:24:23.598Z)

```

1      # Experiment harness ? A/B + shadow eval for rally-detection quality (no-regression by
design)
2
3      **Status:** **P1 IMPLEMENTED** (`backend/eval/experiment.py` +
`tests/test_experiment.py`, shipped
4      2026-06-13) ? the `Variant` layer, `compare()` (labeled A/B, LOVO-honest),
`compare_unlabeled()`
5      (SxS on NEW unlabeled footage), and the leaderboard. **P2?P4 remain owner-gated.**
Phases below are
6      sequenced but each is a separate PR.
7
8      > **Companion docs:** the cues this experiments *on* are in
9      > [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); the scoring mechanics are
10     > [EVALUATION.md](EVALUATION.md); the no-regression gate + golden labels +
`baseline.json` come from
11     > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md); the modular-cue architecture is
**ADR-007**
12     > ([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md)); the distilled
classifier is
13     > **ADR-004**. The 2-layer mental model is
[HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
14
15     ---
16
17     ## 1. Context ? why this, why now
18
19     PR #112 (merged) added the multi-signal fusion design ? shuttle + person + audio cues,
all **default
20     OFF**. The owner's directive for *how* we evolve quality from here:
21
22     > "Evolve the codebase so we can keep finding new ways to improve quality and
experiment with them ?
23     > or add them as a signal ? without suffering any regressions if the idea doesn't work
out. Rolling
24     > back the code would be too dangerous."
25
26     So **experimentation and deployment must be decoupled.** The contract:
27
28     1. Merge experimental code freely ? it's **gated OFF**, so it cannot change production
behavior.
29     2. Test it **offline against golden labels** (most experiments never touch the served
path at all).
30     3. Promote a variant to default **only on measured lift**, through a CI eval gate.
31     4. **"Rollback" = flip a config flag, never `git revert`.** The proven path is a pinned
variant that is
32     always present.
33
34     **The pleasant surprise (Explore audit, 2026-06-13):** ~80% of this harness already
exists and is
35     simply not composed *as a system*. The missing piece is a thin **variant/experiment
layer + the
36     discipline**, not a platform. S4 is the exact code surface so the next session does not
re-discover it.
37
38     ---
39
40     ## 2. The thesis ? separate expensive DETECTION from cheap DECISION
41
42     The expensive, GPU/WSL-bound, risky part is **detection**: "what signals exist per
frame" (shuttle
43     WASB trajectory, person tracks/boxes, audio hits). The cheap, swappable, pure-Python
part is the
44     **decision**: "given those signals, where are the rallies" (windowing + gates + the
classifier).
45

```

```

46     > **Run detection once per video, persist all per-cue signals + per-candidate features,
then re-score
47     > any number of variants OFFLINE ? deterministically, with no GPU and zero production
risk.**
48
49     This is not aspirational ? `distill_local.py` and `calibrate_local.py` already do
exactly this on
50     stored WASB trajectories. The candidate-segments table (`stats` JSON) is the persistence
substrate.
51     ? Most experiments are pure re-scoring of stored data; they never enter the served
pipeline.
52
53     ---
54
55     ## 3. What already exists (the audit ? do NOT rebuild)
56
57     Exact public surface, with `file:line`. **This table is the anti-duplicate-work
artifact** ? start here.
58
59     ### 3.1 Scoring (variant-agnostic ? grades any `List[Interval]`)
60     `backend/eval/metrics.py`
61     - `interval_iou(a, b) -> float` (:15) ? temporal IoU of two `(start,end)` intervals.
62     - `match_intervals(preds, gts, iou_threshold=0.5) -> MatchResult` (:48) ? greedy 1-to-1
IoU matching.
63     - `score(preds, gts, iou_threshold=0.5, match_result=None) -> Score` (:104) ? P/R/F1,
mean IoU, boundary errors; `Score.as_dict()`.
64     - `pr_curve(preds, gts, thresholds=None) -> List[Score]` (:138).
65
66     ### 3.2 A/B + cross-validation primitives (the comparison engine already exists)
67     `backend/eval/calibration.py`
68     - `improvement_report(predict_fn, baseline_config, tuned_config, gts, iou_threshold=0.5)
-> dict` (:148) ? **before/after metrics + %?. This IS the A/B delta.**
69     - `leave_one_video_out(videos, configs, metric="f1", iou_threshold=0.5) -> dict` (:113)
? **LOVO: pick on other videos, score on held-out video (honest cross-video).**
70     - `cross_validate(predict_fn, configs, gts, k=5, metric="recall", iou_threshold=0.5) ->
dict` (:72) ? within-video k-fold overfit guard.
71     - `select_best(predict_fn, configs, gts, metric="f1", iou_threshold=0.5) -> (Config,
float)` (:61).
72     - `evaluate(preds, gts, iou_threshold=0.5) -> Score` (:41); `kfold_indices(n, k)` (:46).
73     - LOVO video dict shape: `{"name": str, "predict_fn": PredictFn, "gts": [Interval]}`.
74
75     ### 3.3 No-regression gate + baselines (the promotion gate already exists)
76     `backend/eval/regression.py`
77     - `regress(db, video_id, ground_truth, stage="final", iou_threshold=0.5, detector=None,
tolerance=DEFAULT_TOLERANCE, baseline_path=DEFAULT_BASELINE_PATH) -> RegressionResult` (:104) ?
score stored preds vs GT, flag if P **or** R drops > tolerance.
78     - `regress_with_provider(...)` (:160) ? same, fetching GT via the annotation provider.
79     - `save_baseline(video_id, stage, precision, recall, f1, ..., gt_fingerprint=None)`
(:80) . `load_baselines(path) -> dict` (:68).
80     - `RegressionResult` (:43): `regressed`, `reasons`, `gt_hash`, baseline P/R,
`tolerance`, `has_baseline`; `as_dict()`.
81     - Default baseline file: `eval_baselines/regression_baseline.json` (:33).
82
83     `run_regression.py` ? CI-gatable CLI, `main(argv)` (:52). **Exit codes: 0 ok / 1
regression / 2 no-DB / 3 no-baseline.** Flags: `--video-id --tier --annotations
--stage{candidates,final} --detector --iou --tolerance --baseline --update-baseline --allow-
missing-baseline --json`.
84
85     ### 3.4 Pinnable model artifact
86     `backend/eval/classifier.py` ? `LogisticRegression` (pure numpy, L2):
`fit(X,y,feature_names=,class_weight="balanced")` (:40), `predict_proba` (:75),
`predict(threshold=0.5)` (:81), `to_dict`/`from_dict` (:86/:97), `save`/`load` (JSON)
(:107/:111). **A trained model = one JSON file ? a pinnable, hashable variant artifact.**
87
88     ### 3.5 Offline re-scoring substrate (persist once, re-score forever)

```

```

89     `backend/infrastructure/database.py`
90     - `add_candidate_segment(video_id, detector, start_time, end_time, chunk_index=None,
confidence=None, stats=None, detection_params=None) -> Optional[int]` (:240) ?
`stats`/`detection_params` stored as JSON; `candidate_segments` table cols incl. `stats`,
`detection_params`, `human_label`, `confirmed_segment_id` (schema :97).
91     - `query_candidate_segments(video_id, detector=None, only_unlabeled=False) ->
List[dict]` (:277) ? returns rows with deserialized `stats`.
92
93     ### 3.6 Label assignment + feature glue
94     `backend/eval/training.py`
95     - `label_candidates(candidate_intervals, gt_intervals, iou_threshold=0.5) -> List[int]`
(:50) ? y-labels by IoU match to golden (same matcher as the evaluator).
96     - `build_training_set(candidate_rows, gt_intervals, iou_threshold=0.5,
feature_fn=extract_yolo_features) -> (X, y, intervals)` (:64) ? pluggable `feature_fn`.
97
98     ### 3.7 Existing LOVO drivers to mirror (not duplicate)
99     - `backend/eval/distill_local.py` ? `run(specs, iou=0.5, threshold=0.5, model_out=None)`
(:98), CLI `python -m backend.eval.distill_local --manifest videos.json --model-out ...`;
`FEATURE_NAMES = [duration, peak_velocity, mean_velocity, active_ratio, net_crossings]` (:40);
already does LOVO over Gemini silver labels.
100    - `backend/eval/calibrate_local.py` ? `run(specs, iou=0.5, metric="f1")` (:77);
`LocalConfig = (velocity_thresh, merge_gap, min_window_duration, min_crossings)`; grid + LOVO.
101    - `backend/eval/calibrate_wasb.py` ? `run(...)` , `load_golden_gts()`;
`backend/eval/gt_loader.py::load_gt_intervals(csv_path, *, strict=True)` is THE canonical golden
reader.
102
103    ### 3.8 Registries + config knobs (variant selection)
104    - `backend/pipeline/segmenters/base.py` ? `SegmenterRegistry` (:35), singleton
`segmenter_registry` (:63); registered: `motion`, `gemini`, `yolo_hybrid`, `tracknet_hybrid`,
`wasb_hybrid`.
105    - `backend/annotations/base.py` ? `AnnotationProviderRegistry`, singleton
`annotation_registry`; providers `file`, `auto`.
106    - `backend/config/models.py::IndexingConfig` ? `default_segmenter` (:65, default
`"yolo_hybrid"`), `detector_model` (:72), `detector_score_threshold` (:73),
`yolo_velocity_threshold` (:74), `yolo_frame_skip` (:75), `min_segment_duration` (:68),
`dead_time_merge_gap` (:69), `motion_threshold` (:67). **No `min_crossings` knob yet** (see
fusion P2).
107
108    ### 3.9 Confirmed ABSENT (no duplication risk)
109    Explore found **no** existing experiment / variant / shadow / A-B *machinery* ? only a
stray "A/B test"
110    aspiration comment in `backend/pipeline/detectors/base.py` and an "experiment E1" note
in
111    `AUDIO_RALLY_DETECTION_PLAN.md`. The `regress()` + baseline path is the canonical
comparison mechanism.
112
113    ---
114
115    ## 4. The variant abstraction (the one thin thing to add)
116
117    A **`Variant` = a declarative recipe**, not a code branch:
118
119    ```
120    Variant = (segmenter_name, IndexingConfig overrides, enabled cues + per-cue params,
classifier model path/hash)
121    ```
122
123    JSON-serializable; selected via the existing `segmenter_registry` +
`IndexingConfig.default_segmenter`.
124    **Control** = a *pinned, frozen* variant (recipe + model hash) that is always registered
and is the
125    default. Adding a new cue = a new Variant differing by one flag ? the control recipe is
untouched.
126
127    ---

```

```

128
129     ## 5. Three experiment modes
130
131     ### 5.1 Offline A/B (primary, label-driven, zero production risk)
132     `compare(videos, variant_A, variant_B)` ? a thin driver composing existing
functions:
133     `query_candidate_segments()` (load persisted features) ? re-score each variant ?
`calibration.improvement_report()`
134     + `calibration.leave_one_video_out()` + `metrics.score()`, graded vs golden
(`gt_loader.load_gt_intervals`).
135     Reports per-video and LOVO-honest aggregate IoU/P/R/F1 deltas. This is the A/B
test, and it needs
136     almost no new scoring code ? only the glue.
137
138     Honest reporting (default ON, additive): `compare(..., honest_stats=True)` also
attaches a "honest"
139     block ? per-metric bootstrap CIs, a paired ?F1 CI + exact sign test (C1), and
F1@k +
140     segment-count ratio (C4) ? alongside the bare-mean fields (existing output
unchanged; `honest_stats=False`
141     restores the legacy shape). A bare LOVO mean at n?6 is not promotable on its own; the
lift is real when the
142     ?F1 CI clears 0 and the sign test agrees. Wiring of the course corrections in
143     [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) §13/C1+C4 (helpers:
`backend/eval/eval_stats.py`,
144     `segmentation_metrics.py`); `record_experiment` carries the honest ?F1 into the
leaderboard.
145
146     ### 5.2 Shadow (for cues that touch detection, e.g. the person detector)
147     The new cue runs and persists its signal into the candidate
`stats`/`detection_params`, but the
148     served decision stays control. Offline you then ask "what would variant B have
scored?" via §5.1 ?
149     B never affects output until it wins. This is how a detection-touching change (not just
a re-scoring
150     change) earns its way in without risk.
151
152     ### 5.3 Promotion gate (no-regression, CI-enforced)
153     `regression.regress()` + `baseline.json` + `run_regression.py` exit codes. A variant
becomes the new
154     default only if it doesn't regress the golden set beyond `tolerance` (exit 1
blocks). Control is
155     pinned ? rollback = flip `default_segmenter`/variant config, never a code revert.
156
157     ---
158
159     ## 6. No-regression guarantees (the heart of the owner's ask)
160
161     - New cues/signals default OFF ? per-cue enable flag in `IndexingConfig`; merged
code can't change
162     behavior until explicitly enabled.
163     - Control variant pinned ? frozen recipe + classifier model hash, always registered
& default.
164     - Promotion only via the eval gate ? `run_regression.py` in CI (exit 1 blocks); no
silent default change.
165     - Experiments are records ? variant-spec hash ? per-video + LOVO scores + label-set
hash + timestamp,
166     persisted to a small JSON experiment leaderboard (generalizes `baseline.json`).
Answers "which
167     signals ever helped, on which video slices," and makes every result reproducible.
168     - Decoupled events ? "merge an experiment" and "change the default" are separate,
independently
169     revertible actions.
170
171     ---

```

```

172
173     ## 7. What to build later (gap list ? thin, additive, owner-gated)
174
175     1. ~~~`backend/eval/experiment.py`~~ ? a `Variant` dataclass + `compare(videos, a, b)`
composing the §3
176         functions (no new scoring math).~~ **DONE (2026-06-13).** Shipped `Variant` (+ pinned
`CONTROL`),
177         `compare` (labeled A/B with LOVO-honest learned-variant training, mirroring
`distill_local`),
178         `compare_unlabeled` (the **SxS-on-new-videos** mode ? agreed / A-only / B-only via
`match_intervals`,
179         no GT), `record_experiment` (? `eval_baselines/experiment_leaderboard.json`), and
`load_video_candidates`
180         (the `query_candidate_segments` bridge). JSON variant recipes first as recommended; a
`VariantRegistry`
181         only if the count grows. 19 tests incl. the no-regression invariant.
182     2. **Persist per-cue features** into candidate `stats` (person: `players_in_court`,
`two_sided_motion`,
183         `in_court_ratio`, `shuttle_player_colocation`; shuttle-state: `net_crossings`,
`direction_reversals`;
184         audio: hit times) so any future variant re-scores offline ? ties to
`MULTI_SIGNAL_FUSION_PLAN.md` §3.2/§4.
185     3. **Per-cue enable flags** in `IndexingConfig` (default `False`); `default_segmenter`
already exists.
186         Add the `min_crossings` knob (fusion P2) here too.
187     4. **CI promotion lane** using `run_regression.py` against the golden `baseline.json`;
write the
188         experiment-leaderboard JSON.
189
190     ---
191
192     ## 8. Phasing (owner-gated; ONE PR per slice, off `master`)
193
194     - **P0 ? this doc**, indexed in `docs/README.md`, cross-linked from
`MULTI_SIGNAL_FUSION_PLAN.md` and
195         `GOLDEN_SET_VENDORING.md`. *(this deliverable; docs-only, no code.)*
196     - **P1 ? `experiment.py` `compare()`** ? **DONE (2026-06-13):** thin wrapper over
`calibration` +
197         `metrics` + `query_candidate_segments`; pure-logic, fully tested in this dev container
(`compare`,
198         `compare_unlabeled`, `record_experiment`, the no-regression invariant).
`compare_unlabeled` adds the
199         SxS read on **new unlabeled footage** the owner asked for (run control + candidate on
a fresh video,
200         diff where they agree/diverge ? two reels via the existing `VideoCompiler` is the GPU-
side follow-on).
201     - **P2 ? per-cue enable flags + feature persistence** into candidate `stats` (default
OFF) ? couples
202         with fusion P1/P2.
203     - **P3 ? CI promotion gate**: `run_regression.py` lane on the golden `baseline.json`;
experiment-leaderboard JSON.
204
205     ---
206
207     ## 9. Scope guardrail
208
209     **Single-tenant / local-first** (per `PLATFORM_ARCHITECTURE.md`): a JSON experiment
registry + the
210     existing harness ? explicitly **NOT** MLflow / W&B / a service. Keep it the size of
`baseline.json`.
211
212     ---
213
214     ## 10. Verification
215

```

```

216 - **Offline, runnable in this dev container (no GPU/torch needed for the glue):** unit
test that
217 `compare()` is deterministic on synthetic candidate rows; a **no-regression invariant
test** asserting
218 "all new-cue flags OFF ? predictions byte-identical to control"; a CI-gate test
asserting
219 `run_regression.py` exit codes (0 ok / 1 regress / 3 no-baseline).
220 - **Machine-side (owner, GPU/WSL):** persist candidates once on a real clip, run
`compare(control,
221 fusion)` LOVO, confirm the reported lift matches a manual `metrics.score()` spot-
check.
222 - **Docs (P0):** links resolve; indexed in `docs/README.md`; cross-linked from
223 `MULTI_SIGNAL_FUSION_PLAN.md` and `GOLDEN_SET_VENDORING.md`.
224 </content>
225

```

10. user (2026-06-14T08:24:24.242Z)

```

1 # Roora Annotation Guidelines ? Rally Golden Set
2
3 **Version: v8 (2026-06-13).** Supersedes the older Drive copies (v3/v5/v6/v7) **and**
the internal
4 "DRAFT v2" label ? those numbers never matched each other; **v8 is the single label to
use with
5 Roora going forward.** Content is otherwise the finalized 2026-06-12 guideline plus the
6 fusion-driven additions marked ***? ADDED (fusion)** below.
7
8 > **?? Canonical home / provenance.** The source of truth for this guideline is the
sibling
9 > **`sports-data-collector`** repo: `docs/annotation/` (`ROORA_BRIEF.md` +
`ANNOTATION_GUIDE.md` +
10 > intake-manifest & NDA templates), generated from `sports-data-collector@ca61405`, and
ingested by
11 > `python -m src.cli.curate import-local`. **This file is a mirror kept here** because
the
12 > multi-signal-fusion features that motivate the new fields (court calibration,
singles/doubles) live
13 > in *this* repo ? see [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md) and
14 > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) (ADR-006). **Any change here must
be synced back
15 > to the collector's `docs/annotation/` before it goes to Roora.** The `? ADDED
(fusion)` callouts are
16 > exactly the deltas to port.
17 >
18 > **`TODO(avidullu)` markers are preserved** ? they are owner-only items (Drive links,
intake
19 > manifest, NDA counsel review) and must be filled before sending to Roora. Search for
`TODO(avidullu)`.
20
21 Brief (vendor spec) + Illustrated Guide (rater how-to), in one doc.
22
23 ---
24
25 # PART A ? Roora Annotation Brief (vendor-facing spec)
26
27 **Project:** racket-sport rally segmentation golden / regression dataset.
28 **Sports in scope:** Badminton, Pickleball, Table Tennis (each in singles AND doubles).
29
30 This brief is the vendor-facing spec. The annotation deliverable is a per-rally CSV that
imports
31 directly via `python -m src.cli.curate import-local`.
32
33 ## 0. What we need in one sentence
34
35 For each provided match video, produce ONE CSV file listing EVERY rally with its start

```



```

time, end
36     time, number of shots, and how it ended ? EXCLUDING all dead time.
37
38     ## 1. Deliverable format (canonical)
39
40     One CSV per video (UTF-8), named exactly `<video_id>.csv` (we supply the `video_id`).
One row per
41     rally, chronological, header row required.
42
43     **Columns:**
44
45     | Column | Type / values | Meaning |
46     |---|---|---|
47     | `rally_number` | integer, 1-based, chronological ? no gaps, no repeats | rally index |
48     | `start_mmss` | `m:ss.mmm` (e.g. `1:12.40`) | rater-entered serve-contact time |
49     | `end_mmss` | `m:ss.mmm` | rater-entered ball/shuttle-dead time |
50     | `start_time` | decimal **seconds** (e.g. `72.40`) ? **AUTO-FILLED** from `start_mmss`
| machine time |
51     | `end_time` | decimal **seconds** ? AUTO-FILLED from `end_mmss` | machine time |
52     | `shots_count` | integer | total racket/paddle/bat contacts (serve = shot 1) |
53     | `ending_reason` | `winner` \| `forced_error` \| `unforced_error` \| `service_fault` \|
`let` \| `other` | WHY it ended |
54     | `last_shot_outcome` | `in` \| `net` \| `out` \| `n_a` | WHERE the last shot landed
(`n_a` for service_fault/let) |
55     | `score_before` | optional, `"A-B" ? blank if not confidently readable | score before
|
56     | `score_after` | optional, `"A-B" | score after |
57     | `notes` | optional free text ? **REQUIRED when `ending_reason = other`** | edge-case
note |
58
59     **Times (mm:ss ? decimal seconds):** raters type only `start_mmss` / `end_mmss`. The
template Google
60     Sheet fills `start_time` / `end_time` with this formula (shown for `start_time`, row 2):
61
62     ```
63     =IF(B2="", "", IFERROR(VALUE(LEFT(B2,FIND(":",B2)-1))*60 +
VALUE(MID(B2,FIND(":",B2)+1,20)), VALUE(B2)))
64     ```
65
66     On "Download as CSV" the computed decimal seconds export. (If your tool already shows
decimal
67     seconds, type those into `start_time` / `end_time` and leave the mm:ss columns blank.)
68
69     **Example (badminton):**
70
71     ```
72     rally_number,start_mmss,end_mmss,start_time,end_time,shots_count,ending_reason,last_shot
_outcome,score_before,score_after,notes
73     1,0:12.50,0:25.00,12.50,25.00,9,winner,in,0-0,1-0,
74     2,0:45.00,0:47.30,45.00,47.30,1,service_fault,n_a,1-0,1-1,serve into net
75     3,1:01.20,1:18.85,61.20,78.85,14,unforced_error,out,1-1,2-1,final clear sailed long (no
pressure)
76     ```
77
78     **Critical format rules:**
79     - `start_time` / `end_time` that reach us MUST be decimal seconds (the formula
guarantees this).
80     - Times are relative to the **START OF THE FILE WE PROVIDE**. Do NOT trim, clip, or re-
encode the
81     video ? our pipeline identifies a video by its file checksum. (The file we share may
be an
82     **annotation-optimized copy** of the original ? lighter and faster to frame-step, but
83     frame-for-frame identical. Treat it exactly like the original: annotate it as-is.)
84     - `end_time > start_time` for every row.
85     - No overlaps: `end_time[N] <= start_time[N+1]`.

```

```

86      - Everything between one rally's `end_time` and the next rally's `start_time` is DEAD
TIME and gets
87      NO row.
88
89      ## 2. The three non-negotiable rules (full detail + pictures in the Guide)
90
91      1. **IGNORE ALL DEAD TIME.** Only label active play.
92      2. **INCLUDE EVERY RALLY ? EXCLUDE NONE.** Every served point, including service faults,
122 shot
93      rallies, and lets/replays. A missed rally is the worst error in this project.
94      - **EXCEPTION:** warm-up / knock-up is NOT a rally ? pre-match and between-games
warm-up looks like
95      rallying but isn't scored play, so don't label it. Tell it apart by: no real serve;
cooperative
96      (not competitive) hitting; the score not progressing; often several shuttles/balls
in use (full
97      how-to in the Guide, Part 2). When unsure if the match has started, flag it.
98      - **SECOND EXCEPTION:** a rally already in progress when the file starts (or still in
progress when
99      it ends) gets NO row ? its true start (or end) is outside the footage and must
never be guessed
100     or extrapolated. Note it in the delivery instead.
101     3. **A SHOT = ONE CONTACT** with the racket/paddle/bat. Serve = shot 1. Bounces and net-
clips are not shots.
102
103     ## 3. Scope & phasing
104
105     - **Phase A ? Pilot (this engagement):** 3 videos (1 badminton + 1 pickleball + 1 table
tennis), each
106     < 3 min with ~10?12 rallies. This is the paid calibration round; we review jointly and
lock the
107     Guide before scale-up.
108     - **Phase B ? Scale (after a successful pilot):** ~12?15 videos PER SPORT (~36?45
total), each ~30 min.
109     Same format, rules, and template.
110
111     Doubles are included across the program. Rally rules are identical to singles; the
badminton- and
112     pickleball-specific doubles notes are in the Guide.
113
114     ## 4. Quality bar (acceptance criteria)
115
116     Automated validation (no overlaps, positive durations, schema-valid) runs on every file,
then a ~20%
117     human spot-check. A file is returned for rework (in scope) if it misses:
118
119     - **Completeness:** every rally present; zero missed rallies; zero dead-time rows.
120     - **Boundaries:** `start_time` / `end_time` within **±0.3 s** of true serve-contact /
landing.
121     - **Boundaries that all sit on a frame-sampling grid (e.g. every 30th frame)
STRUCTURALLY FAIL this
122     bar and are detected automatically ? the whole file is returned for rework. See §7
for the
123     required method.
124     - **`shots_count`: exact for rallies ? 15 shots; ±1 allowed for longer/faster rallies.
125     - **`ending_reason`: one of the six allowed values, per the Guide's decision tree.
126     - **Format:** decimal seconds present, valid columns, non-overlapping, chronological.
127
128     > **? ADDED (fusion) ? court calibration as a day-1 quality lever.** The owner's
pipeline produced a
129     > bad highlight where a held shuttle was mislabeled as a rally. The boundary definition
($ Part 3:
130     > start = serve **contact**, not the toss/hold) already excludes that ? so faithful
boundaries are
131     > themselves a quality checkpoint. The one NEW field we need (next section) is **court

```

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calibration**,
132     > which lets the detector reason about court occupancy and net-crossings.
133
134     ## A. Court calibration & intake manifest    ? ADDED (fusion)
135
136     > **? ADDED (fusion).** *Why:* the multi-signal detector
137     > ([MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md)) needs to know **where the
court is** and
138     > **where the net is** to (a) ignore people outside the court, (b) require ?2 players on
court for a
139     > rally, and (c) count shuttle net-crossings. This is a **one-time, per-video**
annotation ? cheap,
140     > high-leverage, and **required in deliverables from day 1.**
141
142     **Per-video, recorded once in the intake manifest (NOT in the per-rally CSV):**
143
144     | Manifest column | Type | Meaning |
145     |---|---|---|
146     | `video_id` | string | (existing) we supply it |
147     | `fps` | number | (existing) we supply it |
148     | `court_corners` | 4 points `(x,y)` in pixels, clockwise from the near-left corner |
the playable court rectangle |
149     | `net_line` | 2 endpoints `(x,y)` in pixels | the net, left post ? right post |
150     | `singles_or_doubles` | `singles` \ | `doubles` | sets the expected on-court player
count (2 vs 4) |
151
152     - Pick **one clear frame** with the empty/visible court and click the 4 corners + 2 net
endpoints.
153     - If the camera does not move for the whole video (the normal case), one calibration
covers the file.
154     If the camera is moved mid-file, flag it ? we'll split the file.
155     - Worked example (badminton, 1920x1080):
`court_corners=[(360,980),(1560,980),(1330,300),(590,300)]`,
156     `net_line=[(470,640),(1450,640)]`, `singles_or_doubles=singles`.
157
158     *(After the first batch we may add more per-shot fields ? see the $B appendix ? and
renegotiate price.)*
159
160     ## 5. Per-sport quick reference (full version + examples in the Guide)
161
162     - **Badminton (singles & doubles)** ? start: racket?shuttle serve contact . end: shuttle
floors /
163     fault . shot: each racket?shuttle contact.
164     - **Pickleball (singles & doubles)** ? start: paddle?ball serve contact . end: double
bounce / out /
165     net / fault . shot: each paddle?ball contact (floor bounces are NOT shots; two-bounce
rule is normal play).
166     - **Table tennis** ? start: racket?ball serve strike (not the toss) . end: failed legal
return . shot:
167     each racket?ball contact (table bounces are NOT shots). Net-cord serve = let.
168
169     ## 6. `ending_reason` values (decision tree in the Guide)
170
171     This vocabulary attributes WHY the point ended; it is shared across all net-separated
sports
172     (badminton, pickleball, table tennis, tennis). Record WHERE the last shot physically
landed in the
173     separate `last_shot_outcome` column (`in` | `net` | `out` | `n_a`) ? e.g. a forced error
that flew
174     long is `ending_reason=forced_error, last_shot_outcome=out`.
175
176     - `winner` ? last shot landed IN and the opponent made no legal contact (a clean
winner).
177     - `forced_error` ? the losing player erred (netted / out / missed) UNDER PRESSURE ?
stretched,

```

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178     wrong-footed, or rushed by the opponent's strong shot.
179     - `unforced_error` ? the losing player erred on a ROUTINE, UNPRESSURED ball they had
time and position
180         to make.
181     - `service_fault` ? the serve itself failed / was illegal.
182     - `let` ? the point was replayed.
183     - `other` ? anything else / cannot tell. MUST add a note.
184
185     ## 7. Capture method (required for boundary frames)
186
187     Use any tool that shows current time (`mm:ss.mmm` or seconds, or frame + known fps) and
can step one
188     frame forward/back: a frame-steppable player (VLC frame-step) or a timeline tool (Label
Studio /
189     CVAT). The deliverable is the CSV in the schema above ? or, if you work in CVAT, the
CVAT export as-is
190     (we recompute each rally's time from its box frame range; seconds = `frame_number /
fps`, fps provided
191     per video).
192
193     **Boundary placement is the non-negotiable part:**
194     - Set up the CVAT task at **FRAME STEP 1** on the file we provide (it is encoded to make
this fast ?
195     frame-stepping and back-scrubbing stay snappy).
196     - The FIRST box of every rally goes on the EXACT serve-contact frame; the LAST box on
the EXACT
197     shuttle/ball-dead frame. Frame-step to find them.
198     - Boxes IN BETWEEN may stay sparse (e.g. every 30th frame) ? only the two boundary
frames must be exact.
199     - Self-check before submitting: every start/end within  $\pm 0.3$  s of the true frame (Guide).
200
201     ## 8. Logistics & contacts
202
203     - **TODO(avidullu):** paste the Google Drive folder link where source videos are shared
with Roora.
204     - Per-video fps + `video_id` (+ the calibration columns from $A): provided in the intake
manifest
205     (template `INTAKE_MANIFEST_TEMPLATE.csv`). **TODO(avidullu):** fill the manifest (one
row per video)
206     and share its link.
207     - **TODO(avidullu):** paste the Drive folder link where Roora uploads completed
`<video_id>.csv` files.
208     - Blank annotation template Sheet (mm:ss ? seconds formula from $1 pre-filled):
209     **TODO(avidullu):** create it and paste the link.
210     - **TODO(avidullu):** add the escalation channel (email / WhatsApp group) for edge-case
questions.
211     - NDA before any footage is shared ? template `NDA_TEMPLATE.md`. **TODO(avidullu):**
have counsel
212     review, fill the blanks, and have Roora sign.
213     - Roora contact on file: info@rooragrp.com (HSR Layout, Bangalore).
214
215     > **? ADDED (fusion) ? data-handling guardrails (put in the brief / contract):**
216     > - **Exclude minors entirely.** If footage contains children, that video does **not**
enter the
217     > labeled pool ? do not annotate it; flag it back to us.
218     > - **No biometric / identity labeling.** Refer to players only by **per-video
pseudonyms** (e.g.
219     > "near side" / "far side") ? never a real name, and never a cross-video person ID. We
do **not**
220     > collect gait or biometric annotations.
221     > - **No pose / skeleton / keypoint annotation** is requested.
222     > - **Work-for-hire IP assignment** (atop the NDA): the labels Roora produces are
assigned to us as
223     > owned work product.
224

```

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225    ## 9. Pricing request (please quote)
226
227    1. Price per rally and per minute of footage.
228    2. Pilot price (3 short videos, ~30?36 rallies) as a paid calibration round.
229    3. Indicative scale price for ~36?45 longer videos.
230    4. Turnaround for the pilot and throughput (rallies/day) at scale.
231    5. QA process and rework policy.
232    6. Whether annotators have racket-sport familiarity (helpful, not required).
233    7. Confirmation you can deliver the CSV schema above **(incl. the $A intake-manifest
calibration columns).**
234
235    ---
236
237    # PART B ? Illustrated Annotation Guide (rater how-to)
238
239    **Sports:** Badminton, Pickleball, Table Tennis (singles AND doubles).
240
241    > **Illustrations:** boxes marked `[SCREENSHOT: ...]` are placeholders.
242    > **TODO(avidullu):** replace every `[SCREENSHOT: ...]` placeholder with a real frame
grab from the
243    > pilot videos before sending this guide to Roora.
244
245    ## Part 1 ? What you are producing
246
247    One CSV per video (from the template Sheet), one row per rally:
248    `rally_number, start_mmss, end_mmss, start_time, end_time, shots_count, ending_reason,
last_shot_outcome, score_before, score_after, notes`.
249
250    You type: `rally_number`, `start_mmss`, `end_mmss`, `shots_count`, `ending_reason`,
`last_shot_outcome`
251    (+ optional score/notes). The `start_time` / `end_time` decimal-second columns FILL
THEMSELVES via a
252    formula already in the template.
253
254    For every rally you record:
255    - When it started (serve contact) ? `start_mmss` (`m:ss.mmm`, e.g. `1:12.40`)
256    - When it ended (shuttle/ball dead) ? `end_mmss`
257    - How many shots happened ? `shots_count`
258    - Why it ended ? `ending_reason` . Where the last shot landed ? `last_shot_outcome`
259
260    Everything else (gaps between rallies) you IGNORE.
261
262    ## Part 2 ? The three golden rules
263
264    **RULE 1: IGNORE ALL DEAD TIME.** A rally is CONTINUOUS active play. The moment play
stops, the rally
265    is over. The time until the next serve is DEAD TIME and is never labeled.
266
267    ...
268
269    DEAD          RALLY 1          DEAD          RALLY 2          DEAD
(ignore) |=====| (ignore) |=====| (ignore)
270
271    ^           ^           ^           ^
272    serve      shuttle     serve      shuttle
273    contact    lands       contact    lands
274    =start_1   =end_1      =start_2   =end_2
275
276    ...
277
278    Dead time includes: walking back to serve, picking up the shuttle/ball, wiping sweat,
bouncing the
279    ball before serving, the score being announced, replays/slow-mo, crowd or coach shots,
changeovers,
280    towel/medical timeouts, and PRE-MATCH / BETWEEN-GAMES WARM-UP (knock-up) ? see the Rule
2 caveat below.
281
282    **RULE 2: INCLUDE EVERY RALLY ? EVEN THE UGLY ONES.** Every served point gets a row,

```

```

including:
281     - Service faults (serve into net/out/illegal) ? usually `shots_count = 1`,
`ending_reason = service_fault`.
282     - Very short rallies (serve + one error).
283     - Lets / replays ? `ending_reason = let`; the replayed point is its OWN next row.
284
285     A missed rally is the worst error in this project ? the model must learn that short and
failed rallies
286     are still rallies.
287
288     > ?? BUT WARM-UP / KNOCK-UP IS NOT A RALLY ? a common trap, because Rule 2 says
"include
289     > everything." Pre-match and between-games warm-up LOOKS like rallying but is not scored
play; do not
290     > label it. How to tell it apart:
291     > - No real serve. Real play starts with a proper serve from the service position
(diagonal,
292     > behind the service line). Warm-up has players feeding cooperatively from mid-court
with no serve.
293     > - Cooperative, not competitive. Gentle hits straight to each other; nobody is
trying to win the
294     > point (no smashes-to-win, no scrambling/diving).
295     > - Score isn't progressing. No umpire / scoreboard not running / players casually
retrieve and
296     > re-feed; the score doesn't change. Often several shuttles/balls in use at once.
297     > - Per sport: badminton & pickleball ? real play starts with the underhand serve
from the service
298     > court / behind the baseline; table tennis ? real play starts with the tossed, two-
bounce serve.
299     >
300     > When unsure: find the first proper serve where the score starts to progress ? labeling
begins there.
301     > If you genuinely can't tell whether the match has started, flag the clip to us ? don't
guess.
302
303     RULE 3: A SHOT = ONE CONTACT WITH THE RACKET / PADDLE / BAT. Count every time the
shuttle/ball
304     touches a racket, paddle, or bat. The serve is shot 1.
305     - Count: any racket/paddle/bat contact (serve, return, smash, block?).
306     - Don't count: floor bounces, table bounces, net clips, the toss before a serve.
307
308     5-shot example: serve(1) ? return(2) ? hit(3) ? hit(4) ? winner(5) ? `shots_count = 5`,
309     `ending_reason = winner`.
310
311     ## Part 3 ? Where exactly does a rally start and end?
312
313     START (`start_mmss`): the exact frame the server's racket/paddle/bat CONTACTS the
shuttle/ball.
314     Not the toss, not the wind-up ? the contact frame. `[SCREENSHOT: serve-contact
frame]`
315
316     > NOTE (fusion): this is the field that fixes the "held-shuttle" false positive ?
a player
317     > holding/flicking the shuttle before serving is dead time, so it never gets a row.
Faithful
318     > serve-contact boundaries are exactly what the detector is graded against.
319
320     END (`end_mmss`): the exact frame the shuttle/ball becomes DEAD:
321     - Badminton: shuttle touches the floor, or play stops on a fault.
322     - Pickleball: the second floor bounce, or the frame the ball hits the net / lands out.
323     - Table tennis: the second bounce on one side, a non-serve net hit, or off the table.
324
325     `[SCREENSHOT: end frame]`
326
327     If occluded/off-screen, pick the closest clear frame and add a note. Boundary

```

```

tolerance: ±0.3 s
328     (~±9 frames at 30 fps).**
329
330     ## Part 4 ? `ending_reason`: how to choose
331
332     Pick exactly ONE value for how the rally ended. Stop at the first match:
333     1. Did the SERVE itself fail / was illegal (and ended the point)? ? `service_fault`
334     2. Was the point REPLAYED (let / interruption)? ? `let`
335     3. Did the final shot land IN and the opponent make NO legal contact? ? `winner` (a
clean winner)
336     4. Did the losing player make an error (netted / out / missed) UNDER PRESSURE ?
stretched,
337         wrong-footed, or rushed by a strong shot? ? `forced_error`
338     5. Did the losing player make an error on a ROUTINE, UNPRESSURED ball they had the time
and position
339         to make? ? `unforced_error`
340     6. None of the above / cannot tell ? `other` (add a note)
341
342     **FORCED vs UNFORCED ? the one judgment call.** Decide using the opponent's PRECEDING
shot:
343     - `forced_error` = the opponent's shot CAUSED the miss (pace, placement, deception; the
player barely
344         reached it or was pulled out of position).
345     - `unforced_error` = a clean, makeable ball the player missed on their own.
346     - When genuinely unsure between the two, choose `unforced_error` and add a brief note.
347
348     Notes:
349     - A clean winner means NO legal contact by the opponent. If they got a racket/paddle/bat
on it but
350         netted or hit out, that is a `forced_error` / `unforced_error` (their shot), not a
winner.
351     - A serve into the net is `service_fault` (rule 1 beats everything).
352
353     **THEN SET `last_shot_outcome`** ? where the last shot physically went. This is a
SEPARATE,
354     observational column (no judgment): just look at the last shot.
355     - `in` ? landed in (winners; also a put-away the opponent couldn't reach).
356     - `net` ? went into the net.
357     - `out` ? landed outside the court / past the lines.
358     - `n_a` ? not applicable: use for `service_fault` and `let`.
359
360     So each rally gets both: WHY (`ending_reason`) and WHERE (`last_shot_outcome`) ? e.g. a
smash winner =
361     `winner, in`; a clear forced long = `forced_error, out`; a routine ball netted =
`unforced_error, net`.
362
363     ## Part 5 ? Per-sport detail
364
365     ### Badminton
366     - **Serve:** underhand, below the waist, diagonal. Start = racket?shuttle contact.
367     - **Rally end:** shuttle hits the floor, or play stops on a fault.
368     - **Shots:** every racket?shuttle contact; serve = shot 1.
369     - **Service faults** (1-shot rally, `service_fault`): serve into net, serve outside
service court,
370         contact above waist, foot fault.
371
372     Worked example: `[SCREENSHOT: badminton serve]`
373     - Rally 4: serve `1:28.30`, long rally, smash winner, shuttle lands `1:44.10`, 12
contacts.
374         `4, 1:28.30, 1:44.10, (auto), (auto), 12, winner, in, 3-5, 4-5,`
375     - Rally 5: serve `1:57.00` straight into the net.
376         `5, 1:57.00, 1:57.90, (auto), (auto), 1, service_fault, n_a, 4-5, 4-6,`
377
378     **Badminton ? doubles (2 per side).** All rally rules are IDENTICAL to singles:
379     - start = serve contact; end = shuttle down / fault; a SHOT = any racket?shuttle contact

```

```

by EITHER
380     player on a side.
381     - `shots_count` = TOTAL contacts across both teams in the rally. Do not track which
player hit ? just
382     count every contact.
383     - Net exchanges (drives, flat pushes) come fast ? frame-step so you neither miss nor
double-count.
384     - Only the designated receiver returns the serve; if the wrong player returns, the rally
ends on a
385     fault ? still a row (`service_fault` if it's a serve/receiving-position fault, else
`other` + note).
386     - Two teammates may NOT both hit the shuttle in succession (double-hit). Two contacts on
the SAME side
387     back-to-back = a fault ending the rally ? count both contacts, then the rally ends.
388
389     ### Pickleball
390     - **Serve:** underhand, paddle below waist, hit diagonally; must clear the non-volley
zone (the
391     "kitchen"). Start = paddle?ball contact.
392     - **Two-bounce rule:** the serve must bounce once in the receiver's court and the return
must bounce
393     too, before either side may volley. These bounces are NORMAL play, not rally ends, and
bounces are
394     NOT shots.
395     - **Rally end:** ball bounces twice, lands out, hits the net (fails to cross), or a
fault (e.g.
396     volleying in the kitchen).
397     - **Shots:** every paddle?ball contact only; serve = shot 1; floor bounces are not
shots.
398     - **Service faults** (`service_fault`): serve into net, out/long, short in the kitchen,
foot fault.
399
400     Worked example: `[SCREENSHOT: pickleball serve + kitchen line]`
401     - Rally 2: serve `0:33.10`; serve bounces, return bounces, a few volleys; receiver nets
an easy one at
402     `0:41.85`; 6 paddle contacts.
403     `2, 0:33.10, 0:41.85, (auto), (auto), 6, unforced_error, net, 1-0, 2-0, easy ball
netted`
404
405     **Pickleball ? doubles (the standard format, 2 per side).** All rally rules IDENTICAL to
singles:
406     - start = serve contact; a SHOT = any paddle?ball contact by EITHER teammate; floor
bounces still not
407     shots; two-bounce rule still applies.
408     - `shots_count` = TOTAL paddle contacts across both teams; don't track which player.
409     - Serving sequence (FYI only ? you do NOT annotate it): both partners serve before the
serve passes to
410     the opponents, except the very first service turn of the game. The called score has
three numbers.
411     Ignore for annotation; just mark boundaries / shots / ending.
412     - If both teammates contact the ball in succession, that's a fault ? rally ends; count
the contacts,
413     set the ending per the fault.
414
415     ### Table tennis
416     - **Serve:** ball tossed up from an open palm, then struck so it bounces once on each
side. Start = the
417     racket?ball strike (not the toss).
418     - **Rally end:** a player fails a legal return ? ball double-bounces on their side, goes
off the table,
419     or is hit into the net on a non-serve shot.
420     - **Shots:** every racket?ball contact only; serve = shot 1; table bounces are not
shots.
421     - **Let:** a serve that clips the net but is otherwise legal ? the point is replayed.
422     `ending_reason = let`, and the replay is its own next row.

```



```

423 - **Service faults** (`service_fault`): missed/illegal toss, serve into net, serve off
the table.
424 - **Speed warning:** contacts can be < 0.2 s apart ? step frame-by-frame to count shots.
425 - **Doubles** (if a clip appears): partners must hit in STRICT ALTERNATION and serve
diagonally; a
426 missed alternation is a fault that ends the rally. Same rule ? count every racket?ball
contact as a
427 shot; `shots_count` is the team total.
428
429 Worked example: `[SCREENSHOT: table-tennis serve contact]`
430 - Rally 9: serve `4:10.40`; fast 7-contact rally; forehand winner, opponent no contact;
dead `4:14.05`.
431 `9, 4:10.40, 4:14.05, (auto), (auto), 7, winner, in, 6-9, 6-10,`
432 - Rally 10: serve `4:22.10` clips the net (otherwise good) ? let.
433 `10, 4:22.10, 4:23.00, (auto), (auto), 1, let, n_a, 6-10, 6-10, net-cord serve,
replayed`
434
435 ## Part 6 ? Edge cases & FAQ
436
437 - **Two rallies with almost no gap?** Still two rows. `end_time[N] <= start_time[N+1]`,
never overlap.
438 - **Interrupted & replayed?** Label the interrupted attempt as its own row
(`ending_reason=let`), then
439 the replayed point as the next row.
440 - **Serve tossed but caught / re-served (no contact)?** No contact = no rally yet. Start
at the actual
441 serve contact; don't label the aborted toss.
442 - **Opponent reached the ball but netted/hit out?** That's a
`forced_error`/`unforced_error` (their
443 shot), not winner ? forced if your shot pressured them, unforced if it was routine.
444 - **Can't read the scoreboard?** Leave `score_before`/`score_after` blank. Never guess.
445 - **"Let" where the rules say play on?** If play continued, it's NOT a let ? keep one
rally and judge
446 the real ending.
447 - **Warm-up / knock-up / practice points?** Not real rallies ? do not label them. Start
at the first
448 real, scored serve. See the Rule 2 caveat (Part 2). When unsure whether the match has
started, flag
449 the clip ? don't guess.
450 - **Doubles** Same rules ? a shot is any one racket/paddle contact by whichever player
hits it. Never
451 attribute `shots_count` to an individual, just total the contacts.
452 - **Server completely whiffs (no contact)?** Rare. True zero contact ? `shots_count =
0`,
453 `ending_reason = service_fault`, add a note. A graze = 1 contact = 1 shot.
454
455 ## Part 7 ? Self-check before you submit (per video)
456
457 - [ ] `rally_number` is 1,2,3? no gaps or repeats.
458 - [ ] Every `start_time < end_time` (check the auto-filled decimal columns populated).
459 - [ ] No overlaps: `end_time[N] <= start_time[N+1]`.
460 - [ ] Times entered as `m:ss.mmm`; decimal `start_time`/`end_time` auto-filled and look
right.
461 - [ ] No dead-time rows (scan the gaps ? dead time, not missing rallies).
462 - [ ] Every served point present, including service faults and lets.
463 - [ ] `shots_count` looks right (re-step any rally you rushed; doubles = team total).
464 - [ ] `ending_reason` in {winner, forced_error, unforced_error, service_fault, let,
other}; notes
465 filled when `other`.
466 - [ ] `last_shot_outcome` in {in, net, out, n_a} (`n_a` for service_fault/let).
467 - [ ] File downloaded as `.csv`, header row present, UTF-8.
468 - [ ] Did NOT trim / re-encode / rename the source video.
469 - [ ] **? (fusion)** intake-manifest row filled: `court_corners`, `net_line`,
`singles_or_doubles`.
470

```

```

471     ## Part 8 ? Mini glossary
472
473     - Rally ? one continuous passage of play from serve contact to the shuttle/ball
dead.
474     - Shot ? one contact between the shuttle/ball and a racket/paddle/bat (either
teammate in doubles).
475     - Dead time ? any time the shuttle/ball is not in active play. Never labeled.
476     - Service fault ? a failed/illegal serve that ends the point. Still a labeled rally.
477     - Let ? a replayed point. Labeled as its own rally with `ending_reason=let`.
478     - Winner / forced_error / unforced_error ? the final shot was a clean winner
(opponent made no
479     legal contact) / the loser's error was forced by a pressuring shot / the loser's error
was on a
480     routine, unpressured ball.
481     - start_time / end_time ? decimal seconds from the start of the provided file (auto-
filled from mm:ss).
482     - ? court_corners / net_line / singles_or_doubles ? per-video calibration in the
intake manifest ($A).
483
484     ---
485
486     ## Appendix B ? Tier-B extensions (NOT in the day-1 deliverable)    ? ADDED (fusion)
487
488     > Pilot these on the 3-video calibration round, then renegotiate price before scaling.
Do not start
489     > these without explicit instruction ? they are listed so Roora can quote them.
490     >
491     > - Per-shot hit timestamps ? a timestamp for each racket/paddle/bat contact (not
just the
492     > `shots_count` total). Extends the $7 CVAT method (box every contact, not only the
two boundary
493     > frames). Trains/validates the audio hit-detector and shot-cadence features.
494     > - Per-shot type ? for each contact, a label from {serve, clear, drop, smash, net,
drive, lift}.
495     > Needs its own short guideline + an inter-annotator agreement check.
496

```

11. assistant (2026-06-14T08:24:28.166Z)

This is highly informative. The key finding is already crystallizing: shot-type labeling ({serve, clear, drop, smash, net, drive, lift}) is explicitly a **Tier-B extension** in Appendix B, NOT in the day-1 deliverable, and needs its own guideline + inter-annotator agreement check. Let me check the correction loop (the per-user label source the pivot depends on) and the shots_count usage to understand what shot-level signal already exists.

12. user (2026-06-14T08:24:30.159Z)

Found 8 files

```

backend\infrastructure\database.py
backend\eval\score_calibration.py
backend\eval\segmentation_metrics.py
backend\eval\eval_stats.py
backend\tools\export.py
backend\pipeline\segmenters\yolo_hybrid.py
backend\pipeline\segmenters\motion.py
backend\reporting\harvest.py

```

13. user (2026-06-14T08:24:30.404Z)

```

1     # Rally-detection quality iterations ? living log
2
3     Purpose. A running, reverse-chronological log of how we evolve rally-detection
quality: each
4     iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson.

```

```

This is the
5     **live** continuation of the historical trail in
6     [`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the
archive policy
7     (`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this
file stays at
8     `docs/` top level and keeps growing, then its entries graduate to the archive when they
conclude.
9
10    **Methodology (non-negotiable).** Every quality change rides the experiment harness
11    ([EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)): new cues land **flag-gated, default
OFF**;; they're
12    graded **offline against golden labels** (per-video + LOVO-honest); promotion to default
happens **only
13    on measured lift** through the `run_regression.py` gate; **rollback = flip a config
flag, never
14    `git revert`.** Cues are designed in
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); scoring
15    in [EVALUATION.md](EVALUATION.md); the 2-layer model in
16    [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
17
18    **Golden corpus = the 6 labelled matches** in `output/human_lovo_manifest.json` (owner-
confirmed
19    2026-06-13: use all 6 ? `adarsh_avi_singles, kushagra_singles, targetest_doubles,
gbaaddy,
20    testlarge_short, mahadevpura_singles`). The model trained on these is what we run
alongside the existing
21    detector on NEW footage for the side-by-side review.
22
23    ---
24
25    ## Baselines that drive strategy (IoU 0.5 vs human golden labels)
26
27    | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
28    |---|---|---|
29    | Heuristic (velocity windowing) | 0.657 | 0.157 |
30    | Gen-0 owned head (LOVO, n=6) | 0.744 | 0.561 |
31    | Two-stage WASB+Gemini refine | 0.83 | ? |
32    | Gemini wrapper (`gemini-flash-latest`) | 0.93 | 0.66 |
33
34    **Reading:** clean footage is commoditized (serve via Gemini); **multi-court drops the
commodity wrapper
35    to 0.66 ? that is the owned model's ship target.** Baselines tracked in
`eval_baselines/`. The owner's
36    observed false positive ? a player **holding/flicking** the shuttle between points
counted as a rally ?
37    is the precision bug the fusion gates target.
38
39    ---
40
41    ## Current track (live) ? newest first
42
43    ### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio =
promising (not promotable) *(2026-06-14)*
44    The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ?
`output/*_golden_features.json`, the
45    replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats
(C1 ? bootstrap 95% CI + paired sign
46    test; never a bare mean at n=6)**:
47
48    | Variant (honest LOVO, IoU ? 0.5) | F1 | ?F1 vs prior | 95% CI | sign test | CI clears
0 |
49    |---|---|---|---|---|
50    | shuttle-only | 0.307 | ? | ? | ? | ? |
51    | + person (P3) | 0.286 | **?0.021** (?6.7%) | [?0.165, +0.162] | 2W/4L (p=0.688) |

```

```

**No** |
52 | + person + audio (P4 incr.) | 0.366 | **+0.079** (+27.7%) | [?0.054, +0.203] | 4W/2L
(p=0.688) | **No** |
53
54 **Read honestly:** neither cue clears the promotion bar (both CIs span 0, both sign
tests n.s.) at n=6 ? **both stay
55 default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ?
`players_in_court` counts spectators);
56 per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising
lead ? +0.079 / +27.7%, wins 4/6
57 (largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above**
shuttle-only (0.307) ? but n=6 can't
58 confirm it. **Next:** more matches (raise N until the sign test can resolve ~+0.08), and
measure audio on shuttle-only
59 *directly* (without the person handicap). Recorded in
`eval_baselines/experiment_leaderboard.json`. **Negative/uncertain
60 results are kept on purpose ? this is the honest ablation backbone for the paper aim.**
61
62 ### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125
corroboration *(2026-06-14)*
63 **Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The
measured win
64 (?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg
1.68?1.01 ? the
65 [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md))
clears the honest
66 bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the
eval-only Gate A
67 folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the
recall-oriented *proposal*
68 candidate set; flipping the **live served default** stays owner-gated until a real
fusion-path run confirms it
69 (the production fusion segmenter is P2/unverified on real video) ? so this records the
promotion + rationale,
70 not a silent default flip. **Corroborated on real footage:** offline re-score of
GX010125_1080p's *cached*
71 trajectory (no GPU, isolated `output/GX010125_run`) ? 45 candidates ? CONTROL 45
rallies vs **Gate-A 27** ;
72 Gate-A dropped **18 low-crossing windows (~1.7 min)** of likely held-shuttle/feed FPs
(no golden labels for
73 this footage ? divergence, not F1).
74
75 **Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-
indexer/pull/130).** The
76 first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff
artifact. Added
77 additive/default-OFF `Variant.tune_threshold` (operating point chosen on the **train**
fold) + `calibrate`
78 (Platt/isotonic, C2). Validated on the n=6 golden set: **mean F1 0.307 ? 0.423
(+0.116)** , testlarge
79 0.000 ? 0.300, and the learned variant now **beats CONTROL (0.388)**. Threshold-in-LOVO
is the big lever;
80 calibration is available too.
81
82 **Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
83 ([PER_VIDEO_OUTPUT_LAYOUT.md](PER_VIDEO_OUTPUT_LAYOUT.md)) so each video's artifacts are
self-contained.
84
85 ### First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
86 First real use of the harness + C1?C4 helpers, 100% offline (re-scored persisted WASB
trajectories, no GPU).
87 **Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test
**p = 0.031**, CI
88 [+0.042, +0.104], over-segmentation (segment-count ratio) **1.68 ? 1.01**. The window-
level 5-feature logistic

```

```

89     **did not clear the bar** (?0.081, 2W/4L; a threshold/calibration artifact ? *not* the
stronger per-frame
90     prototype). The framework held up: **C1 promoted one variant and refused another; C4
surfaced over-segmentation
91     tIoU hid.** Full record + tables ?
92     [archives/quality-iterations/2026-06-first-ab-experiment/](archives/quality-
iterations/2026-06-first-ab-experiment/README.md).
93     Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure
is the C2 (calibration)
94     + threshold-in-loop task list.
95
96     ### Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF
*(2026-06-14)*
97     **Why.** A multi-agent literature review (folded into
98     [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) §13) confirmed the
framework's *skeleton* is
99     standard prior art (Temporal Action Localization/Segmentation + late/score-level fusion
? cite, don't claim),
100    and flagged that our **measurement** is walking into known small-N traps that would make
"promote only on
101    measured lift" promote noise or hide a real win. Four corrections, shipped as **pure,
additive helpers** (no
102    existing `compare`/LOVO/`metrics.score` behaviour changed ? callers opt in):
103    - **C1 ? never a single mean.** `backend/eval/eval_stats.py`:
`mean_with_ci`/`bootstrap_ci` (deterministic,
104    seeded), `paired_sign_test` (the exact distribution-free "won on enough *videos*?"
test the corpus already
105    needs), `paired_delta_ci` (B?A with a CI that must clear 0). At n?6 a lone LOVO mean
is not trustworthy.
106    - **C2 ? calibrate cue confidence.** `backend/eval/score_calibration.py`: Platt +
isotonic (PAVA) calibrators
107    + `brier_score`/`expected_calibration_error`, so a CNN softmax, an audio onset, and an
integer net-crossing
108    count are comparable before the arbiter and the over-confident cue can't swamp the
rest.
109    - **C3 ? honest stacking.** `eval_stats.out_of_fold_predictions`/`grouped_folds` (leave-
one-group-out by
110    match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no
leakage).
111    - **C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`:
`f1_at_overlaps`
112    (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU),
`segment_count_ratio`. tIoU alone
113    is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler
exists to fix.
114
115    **Status.** Helpers + 24 unit tests landed (pure stdlib, run in the numpy/GPU-free
lanes); ruff + mypy clean.
116    **Not yet wired into the default harness flow** (wiring `compare`/`run_regression.py` to
report CIs + F1@k is
117    the follow-on, owner-gated). The discipline they encode is the methodology for every
future cue measurement ?
118    see §13/C1?C4 for the citations and the remaining (Tier-2/3) corrections.
119
120    ### Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*
121    **Hypothesis.** Adjudicating shuttle-seeded windows with the net-crossing **Gate A**
(`min_crossings?2`)
122    and a person-occupancy **Gate B** raises **precision** on the held-shuttle / warm-up /
one-sided-feed /
123    spectator false positives ? *without* hurting recall, because the shuttle stays the
primary,
124    camera-invariant seed.
125
126    **Scope ? and an honesty boundary from the archive (read this).** The
127    [2026-06-multivideo-lovo](archives/quality-iterations/2026-06-multivideo-

```

```

lovo/README.md) iteration
128     already smoke-tested **court-region detection as a fix for the *multi-court contrast*
confound** and
129     found it **low-headroom**: only 12%14% of dead-time shuttle detections fall *outside*
the foreground
130     court, so a perfect court gate lifts contrast to only ~1.6%2.2× (below the 3.6× "works"
threshold) ? the
131     multi-court problem is **temporal**, and the **learned model**, not a court mask, is its
answer. So Gate B
132     here is **NOT** sold as the multi-court fix. It targets a **different, previously-
untested** failure mode
133     (held-shuttle/warm-up/one-sided/spectator FPS). Two distinct levers; don't conflate
them.
134
135     **Data support for Gate A.** The same archive shows real rallies on the multi-court
matches contain
136     **2 net-crossings 97% of the time (median 5)** while a single service feed crosses
~once ? `min_crossings`2`
137     is a principled, data-backed discriminator (it's the held-shuttle owner-bug killer).
138
139     **Shipping shape.** A NEW `fusion` segmenter, **default OFF** (`default_segmenter`
unchanged); Gate A
140     wired into the live trajectory path behind `indexing.min_crossings` (default 0 = OFF ?
wasb/tracknet
141     byte-identical today); the court mask is an **optional** config polygon (omit ? Gate B
is
142     player-count-only; supply ? masked). Person features (`players_in_court,
two_sided_motion,
143     mean_player_motion, in_court_ratio, shuttle_player_colocation`) persisted into candidate
`stats` for P3's
144     learned fusion. Reuses `PersonDetector` (P1) + `rally_gate` (already existed) ? no
reinvention.
145
146     **Status (2026-06-14): SHIPPED, default-OFF.** `FusionSegmenter`
(`backend/pipeline/segmenters/fusion_hybrid.py`,
147     registry key `fusion_hybrid`) + the pure feature math
(`backend/pipeline/detectors/fusion_features.py`) + two
148     identity hooks on the trajectory engine (`_enrich_candidate_stats`,
`_select_for_handoff`) + `PersonDetector.
149     detections_per_frame` + the `FusionConfig` block. Built design-first (a
`fusion-p2-design` fan-out workflow), then
150     **adversarially reviewed by a 31-agent workflow (25 findings ? 19 confirmed)**: the
headline confirmation is
151     **NO-REGRESSION verified** ? wasb/tracknet twins persist byte-identical stats and hand
off all windows (pinned by
152     the extended equivalence test); default `FusionSegmenter` == its wasb substrate. The
confirmed low/medium findings
153     were fixed (frame-dim guards, `round()`, frame indices, the per-window axis-estimate
redundancy ? a reusable
154     `rally_gate` estimate, honest torch-import docstrings, and 6 new tests closing the
person-detector / Gate-B /
155     substrate / colocation-halo gaps). Suite 441?467 green**; ruff + mypy clean.
156     **Two findings deferred with intent** (reviewer-agreed): person features are computed
eagerly even for
157     Gate-A-pruned windows (a P3 redesign, would change the persisted-superset semantics);
and `fusion.velocity_threshold`
158     must be set equal to the substrate's to A/B against a tuned wasb run (documented in
`FusionConfig`).
159     **Lift still to be measured** offline on the 6 golden videos via
`experiment.compare(CONTROL, fusion)`; the
160     held-shuttle-drop + person-feature values are GPU/owner-side. *(Measured numbers
appended here when run ? no claims before.)*
161
162     ### Person cue extracted to a reusable producer (fusion P1) ?
[115](https://github.com/avidullu/badminton-highlight-indexer/pull/115), 2026-06-13

```

```

163     Lifted the torchvision detector + ByteTrack + velocity windowing out of `yolo_hybrid`
into
164     `backend/pipeline/detectors/person_detector.py::PersonDetector`. **No behavior change**
(proven by the
165     unchanged `process_video` output tests); `yolo_hybrid` now orchestrates only. **Why it's
a quality move:**
166     it makes "add the person cue to fusion" an *add-a-producer*, not a segmenter rewrite.
Suite 440?441 green.
167
168     ### Experiment harness: A/B without rollback risk ?
[114](https://github.com/avidullu/badminton-highlight-indexer/pull/114), 2026-06-13
169     Shipped `backend/eval/experiment.py` ? the thin variant/experiment layer (the audit's
"~80% already
170     exists" held; pure glue over
`metrics`/`calibration`/`classifier`/`training`/`regression.gt_hash`). Gives
171     `Variant` (+ pinned `CONTROL`), `compare()` (labeled A/B, LOVO-honest),
**`compare_unlabeled()`** (the SxS
172     read on NEW unlabeled footage ? agreed / A-only / B-only), and a JSON experiment
leaderboard. **This is
173     the instrument every iteration is measured on.** 19 tests incl. the no-regression
invariant (all cues OFF
174     ? byte-identical to control). Suite 421?440 green.
175
176     ---
177
178     ## Archived predecessors (the historical trail)
179     Completed iterations live under [archives/quality-iterations/](archives/quality-
iterations/README.md):
180     - **`2026-06-initial`** ? seed of the quality phase.
181     - **`2026-06-human-gt-tuning`** ? first eval vs **human** GT on a partial-label video;
partial-label
182     harness + labeled-window methodology; baseline 0.650 ? tuned 0.742 (*fit-on-self*);
found+fixed a
183     negative-start ingestion bug (collector PR #13).
184     - **`2026-06-multivideo-lovo`** ? the **n=6 corpus + honest LOVO**; established the
heuristic is
185     **footage-fragile** (multi-court Kushagra 0.16 / gBaaddy 0.28), the **"contrast"
metric** predicts
186     heuristic F1 monotonically, and a **learned per-frame prototype overtook the heuristic
at n=3 and holds
187     ~+0.10** (n=6: learned 0.583 vs heuristic 0.480), with **AUC 0.83?0.92 decoupled from
contrast**. This
188     is the data-backed green light for the owned model.
189
190     ---
191
192     ## Standing lessons (carry-forward)
193     - **Persist detection once, re-score variants forever.** Expensive/risky = per-frame
detection (GPU/WSL);
194     cheap/swappable = the decision. Most experiments never enter the served pipeline.
195     - **LOVO or it didn't happen.** Score on a held-out *video*, never a held-out window
from the same video.
196     Group by **match**, not file. Sign test needs n ? ~n=10/9-wins for p<0.05.
197     - **The multi-court confound is TEMPORAL, not spatial/loudness** ? three heuristics were
tested and failed
198     for the same reason (foreground spatial gate **?0.04**; audio energy **AUC ~0.6**;
court-region
199     separability **12?14% out-of-region**): the dead-time confusion is rally-vs-non-rally
activity in the
200     *same place and frequency band*. Only the learned model's temporal/structural features
crack it (AUC
201     0.83 on Kushagra/gBaaddy). **Don't re-derive these dead ends.**
202     - **Court-masking ? multi-court fix** (see above) ? it's a **precision** tool for off-
court persons and
203     one-sided feeds, a *different* lever; measure its lift, don't assume it.

```

```

204     - **Net-crossings are a strong rally signal** ? 97% of real rallies cross ?2× (median
205     Gate A leans on this.
206     - **Small-N feature discipline.** Few, scale/translation-invariant features
(counts/ratios, not raw
207     coords) + a logistic regression, not a net ? raw coords memorize this court/camera.
208     - **Calibration, not representation, is the small-N wall** ? learned probabilities
separate (high AUC) but
209     the operating point doesn't transfer with few training videos. More distinct matches
is the highest-ROI
210     unblock.
211     - **Gate A already existed, just wasn't wired to production** ? the cheapest precision
win is plumbing, not
212     new capability. Always check what the eval drivers already do that the live path
doesn't.
213

```

14. assistant (2026-06-14T08:24:36.328Z)

This is exactly the empirical grounding I need. The QUALITY_ITERATIONS log shows the real measured state: even ****rally**** detection at n=6 golden videos is borderline (shuttle-only F1 0.307, learned-with-threshold 0.423), person cue gives NO lift, audio is promising-but-unconfirmed, and the standing lesson is explicit: "Calibration, not representation, is the small-N wall... More distinct matches is the highest-ROI unblock." Let me verify what shot-level signal exists in code today (the `shots\_count` and `direction\_reversals` references) to ground the cold-start claim.

15. user (2026-06-14T08:24:38.331Z)

```

backend\config\models.py:83:    true ? the person-cue features, but it does NOT gate the served
handoff unless a
backend\config\models.py:97:    # Served Gate A: drop windows whose shuttle net-crossings < this
from the AI handoff.
backend\config\models.py:100:   # Served Gate B: when the person cue is on, drop windows with
median in-court players
backend\config\models.py:132:   # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused
? observed live as
backend\config\models.py:208:   # mode). Default None preserves the single-user local 'any
local file' workflow.
backend\main.py:51:# Serves static frontend files directly (now under api/ per approved
structure)
backend\main.py:56:def serve_index():
backend\main.py:61:    return "<h3>Sports Highlight Indexer Server is Running. Please create
index.html in backend/api/static/</h3>"
backend\main.py:95:    server: Optional[str] = None
backend\main.py:239:    """Job worker: drives _execute_processing and records terminal state.
backend\main.py:313:# proxies cap request bodies at ~100 MB ? HOSTING_PLAN $2), are assembled
server-side
backend\main.py:315:# Upload state is in-process by design (this is the single-box alpha): a
server restart
backend\main.py:451:    server-local path)."""
backend\main.py:474:        "server": req.server,
backend\main.py:616:    parser.add_argument("--server", action="store_true", help="Run the
FastAPI local API server")
backend\main.py:617:    parser.add_argument("--port", type=int, default=8000, help="Port to run
the FastAPI server on")
backend\main.py:694:    if args.video_path and not args.server:
backend\main.py:764:        # Start Local Server Mode
backend\main.py:765:    if args.server or not args.video_path:
backend\main.py:766:        print(f"[SERVER] Starting local FastAPI server on
http://localhost:{args.port}...")
backend\features\_init__.py:5:train-time and serve-time features are identical by construction.
Pure numpy/scipy:
backend\features\rally_seq.py:4:harness (training/) and the serving path import THIS module, so
train == serve by

```



```

backend\features\rally_seq.py:6:59.94fps than 30fps) is exactly the train/serve-skew class this
placement prevents.
backend\infrastructure\database.py:68: server TEXT,
backend\infrastructure\database.py:171: row in place; clearing segments is now an
explicit, deliberate operation
backend\infrastructure\database.py:172: (clear_video_data / prune_segments).
backend\infrastructure\database.py:224: def add_play_segment(self, video_id: str, start_time:
float, end_time: float, server: Optional[str] = None, receiver: Optional[str] = None, metadata:
Optional[Dict[str, Any]] = None) -> Optional[int]:
backend\infrastructure\database.py:229: "INSERT INTO play_segments (video_id,
start_time, end_time, duration, server, receiver, metadata) VALUES (?, ?, ?, ?, ?, ?, ?, ?)",
backend\infrastructure\database.py:230: (video_id, start_time, end_time,
end_time - start_time, server, receiver, json.dumps(metadata or {}))
backend\infrastructure\database.py:326: - server: string

```

[Showing results with pagination = limit: 30]

16. user (2026-06-14T08:24:38.549Z)

```

1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
4      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
6      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
14
15     ## Where we are (one paragraph)
16     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
17     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
18     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
19     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
20     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
21     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
22     battery is done** (see table) ? the moat is quantified.
23
24     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
25     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
26     |---|---|---|
27     | Heuristic (velocity windowing) | 0.657 | 0.157 |
28     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
29     | Two-stage WASB+Gemini refine | 0.83 | ? |
30     | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |

```

```

31
32     **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
33     wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
34     the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
35     architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
36     only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
37     Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
38
39     ## Rally-detection quality track (NEW 2026-06-13 ? design merged, owner-gated, NOT
started)
40     **This is the track a desktop/local agent is slated to pick up.** Three design docs
landed (PR #112 =
41     fusion + Roora v8; this session = experiment harness + handoff). All implementation is
**owner-gated**.
42     - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio cues, one
fusion layer) .
43     `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
44     `docs/ROORA_LABELING_GUIDELINES.md` (v8) . `docs/HOW_RALLY_DETECTION_WORKS.md` (mental
model).
45     - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a
rally" bug is
46     **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already
implements the
47     net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills service-
feed/held-shuttle
48     windows, but it is **only called from eval drivers** (`distill_local.py`,
`eval_partial_labels.py`,
49     `export_wasb_segments.py`, `calibrate_local.py`) and there is **no `min_crossings`
knob in
50     `IndexingConfig`** ? so production never applies it. **Wiring Gate A into the live
segmenter is the
51     single cheapest, highest-confidence quality win.**
52     - **Pick-up order (each = ONE PR off `master`, owner approval first):**
53     1. **Fusion P2 first-step ? wire Gate A into production:** add `min_crossings` to
`IndexingConfig`;
54     apply `rally_gate` in `wasb_hybrid`/`trajectory_hybrid` runtime (today it's eval-
only). Pure-logic,
55     testable in this container. *Fixes the observed bug.*
56     2. ? **DONE (2026-06-13) ? Fusion P1 ? extract the person cue:** lifted torchvision
detector + ByteTrack
57     + velocity out of `yolo_hybrid` into
`backend/pipeline/detectors/person_detector.py` (`PersonDetector`);
58     `yolo_hybrid` now orchestrates only + delegates to `self.person` (no behavior
change ? regression-safe,
59     proven by unchanged process_video tests). `tests/test_person_detector.py` added;
person_detector
60     type-clean. **The fusion segmenter (P2) reuses this producer over shuttle-seeded
windows.**
61     3. ? **DONE (2026-06-13) ? Experiment-harness P1 ? `backend/eval/experiment.py`:**
thin wrapper over
62     the EXISTING `calibration.py`, `metrics.py`, `classifier.py`, `training.py`,
`regression.gt_hash`, and
63     `database.query_candidate_segments` (no rebuilt scoring). Shipped `Variant` (+
pinned `CONTROL`),
64     `compare` (labeled A/B, LOVO-honest for learned variants), `compare_unlabeled`
(**SxS on new
65     unlabeled footage** ? agreed/A-only/B-only), `record_experiment` (leaderboard
JSON), and the DB
66     bridge; 19 tests incl. the no-regression invariant. Suite 421?440 green.

```

```

67      4. ? **DONE (2026-06-14) ? Fusion P2 ? the FusionSegmenter:**
`fusion_hybrid.py::FusionSegmenter`
68      (registry `fusion_hybrid`, **default-OFF**) seeds windows from the shuttle
substrate, persists
69      `net_crossings` (Gate A) + optional person features (`fusion_features.py`, pure-
tested) via two
70      identity hooks on the trajectory engine, with optional served Gate A/B
(`fusion.min_crossings` /
71      `min_players`, default 0) that gates the handoff while persisting the full
superset. Court mask =
72      optional `fusion.court_polygon`. Adversarially reviewed (31 agents, 19 confirmed;
NO-REGRESSION
73      verified); suite 441?467 green. Next: **P3** learned fusion (person features ?
harness `compare`
74      LOVO lift on the 6 golden videos; the eager-person-compute optimisation lands
here), then **P4**
75      audio cue via `backend/pipeline/audio/hit_detector.py`.
76      - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue lands **flag-gated, default
OFF**; promote
77      to default **only on measured LOVO lift** through the `run_regression.py` gate (exit
0/1/3); pinned
78      control variant ? **rollback = config flip, never `git revert`**. The harness is ~80%
existing code.
79      - ? Dev container has no GPU/torch/cv2 ? Gate-A wiring, the person-cue extraction logic,
and
80      `experiment.py` are pure-logic/testable here; real-video confirmation is owner-side
(GPU/WSL).
81
82      ## Prioritized next steps
83      1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
every lever below feeds on it.
84      Priority order for new videos: **(a) multi-court badminton** (the target distribution
AND where the ship target
85      lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
0.909; pickleball labels are
86      corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
87      `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
88      label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
re-run the Gen-0 harness.
89      The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
reachable this month.
90      2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
91      reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93
92      (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with
93      paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
94      3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for
**ActionFormer/ASFormer** (MIT, minutes on the
95      2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing
harness; per-video threshold
96      calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
97      4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
98      badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
99      by C3 (provenance multiplies per sport).
100     5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
101     "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
102     (consent at capture; demo served via the Gemini path or human-polish ? the reel

```

```

pattern exists:
103     `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
104     physical-visit approach; reframed product-feedback-first per owner spec):**
105     `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +
106     `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
107     rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
108     capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md`
(G/A/R anchors + **pre-registered
109     decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
110     6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
111     deletion in AI Studio); **rename repo off "badminton"** (owner action,
112     in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
113
114     ## Pointers / artifacts
115     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` . Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) .
116     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` . Quality history:
117     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
118     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
119     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
120     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest
`output/human_lovo_manifest.json`
121     (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` .
122     labels: `...\Collect\Golden Labelled\`.
123     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
124     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
125     with a MISLEADING "credits depleted" 429; ramp gradually.
126     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
127     PLATFORM slice = async job model (single-tenant).
128

```

17. assistant (2026-06-14T08:24:44.020Z)

I now have a comprehensive empirical picture. Let me do one final targeted check: confirm that `shots\_count` exists only as a golden-label CSV column (not derived/detected in code), which is the crux of the shot-taxonomy cold-start argument. The grep showed no shot-type detection code in `backend`. Let me confirm the `shots\_count` is only a label field and check the `direction\_reversals` claim from HOW_RALLY_DETECTION_WORKS (the one detectable shot-proxy signal).

18. user (2026-06-14T08:24:45.940Z)

```

tests\test_calibrate_wasb.py:27:         w.writerow(["rally_number", "start_time", "end_time",
"shots_count"])

```

19. assistant (2026-06-14T08:24:52.961Z)

This confirms the crux empirically: across the entire Python codebase, `shots\_count` appears only as a CSV header column in a test fixture, and there is **zero** `direction\_reversal` / shot-type detection code. The repo has no shot-taxonomy capability at all today ? not even the `direction\_reversals` proxy feature mentioned aspirationally in HOW_RALLY_DETECTION_WORKS.md is implemented. Shot taxonomy is entirely greenfield.

I have everything I need. Let me synthesize the structured analysis.

20. user (2026-06-14T08:26:17.256Z)

Output does not match required schema: root: must have required property 'dimension', root: must have required property 'pros', root: must have required property 'cons', root: must have required property 'findings', root: must have required property 'shapes_future', root: must have required property 'open_questions'

21. user (2026-06-14T08:27:22.576Z)

Structured output provided successfully

22. assistant (2026-06-14T08:27:33.071Z)

I've completed the analysis of the shot-taxonomy feasibility dimension and returned it via StructuredOutput.

Key grounding from the repo (all file paths absolute):

- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\HOW\_RALLY\_DETECTION\_WORKS.md` ? the 2-layer model; shot taxonomy is a notional third layer (direction-reversals ? shots is a "spark," not built).
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\ROORA\_LABELING\_GUIDELINES.md` ? shot-type `{serve, clear, drop, smash, net, drive, lift}` is explicitly **Tier-B / Appendix B**, NOT day-1, and needs its own guideline + inter-annotator-agreement check.
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY\_ITERATIONS.md` ? the honest measured rally numbers (n=6 golden: shuttle-only F1 0.307, learned+threshold 0.423, person = negative lift, audio CI spans 0) and the standing lesson "Calibration, not representation, is the small-N wall."
- `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\EXPERIMENT\_HARNESS.md` ? the LOVO/A-B substrate the pivot relies on (genuinely additive, but designed for rally-level labels).

The load-bearing empirical finding: a whole-codebase grep found `shots\_count` only as a CSV header in one test fixture and **zero** shot-type/direction-reversal detection code ? shot taxonomy is entirely greenfield, the correction loop is rally-level only, and class imbalance (not just small N) makes "heavily optimized by 10-15 videos" realistic for rally/highlight but not for full shot taxonomy. My recommended phasing puts rally+highlight first, a coarse no-new-model shot proxy second, a finer correction UI + Tier-B vendor labels third, and per-user shot calibration only for the high-volume pro archetype.